EXIM BANK

CUSTOMER-CENTRIC: HOLISTIC APPROACH

PROFIT SECURITY

EXECUTIVE SUMMARY

Background

The bank's customer cycle is a critical process that encompasses all aspects of customer interaction, from acquisition to retention. It is important to understand this cycle to identify opportunities to improve customer retention and profitability.

The EXIM Bank's customer cycle scrutinized from the perspectives of profit security and churn prediction. Profit security is concerned with securing the bank's earning by identifying and mitigating risks associated with customer criteria, while churn prediction is the process of identifying customers who are at risk of leaving the bank and taking their business elsewhere.

The importance of profit security is essential for banks to maintain their profitability and financial stability, while churn prediction can help banks identify and retain valuable customers, which can lead to increased revenue and profitability. Ultimately, the bank can identify several strategies **optimising asset allocation** and that it can use to improve profit security and reduce churn.

Contents as per below:

Security

1- Security of Trade and Non-Trade
2- Security Portfolio
3- Security Criteria Prediction

Financing Asset Quality

1- Overall Financing Asset
2- Impaired Ratio
3- Impaired Behaviour
4- Accounts Migration
5- Cycle Diagram
6- Customer Churn Prediction

Problem statement

The bank is facing increasing competition from both traditional and fintech rivals. As a result, it is more important than ever for it to focus on customer retention and profitability. However, many banks **struggle to allocate their asset correctly**, to predict which customers are at risk of churning and to develop strategies to prevent it. This can lead to significant losses in revenue and profits.

A machine learning model is necessary to accurately predict which customers are at risk of churning and recommend interventions to prevent it. This model will be trained on a variety of data, including customer demographics, history, and facility. By predicting churn and intervening early, banks can reduce churn rates, increase profits, and improve customer retention.

Security of Trade and Non-Trade

EXIM Bank's primary line of business is to provide credit facilities to finance and support exports and imports of goods, services, and overseas projects with emphasis on non-traditional markets as well as the provision of export credit insurance services, export financing insurance, overseas investments insurance and guarantee facilities.

Throughout the year, both trade and non-trade customers, as well as its financing asset's client base and cumulative profit payment are **slowly decreasing** while accrued profit slowly increases. Figures below shows the number of accounts, gross financing asset, cumulative accrued profit, cumulative profit payment and profit security ratio by trade and non-trade facility from year 2020 until 2023:

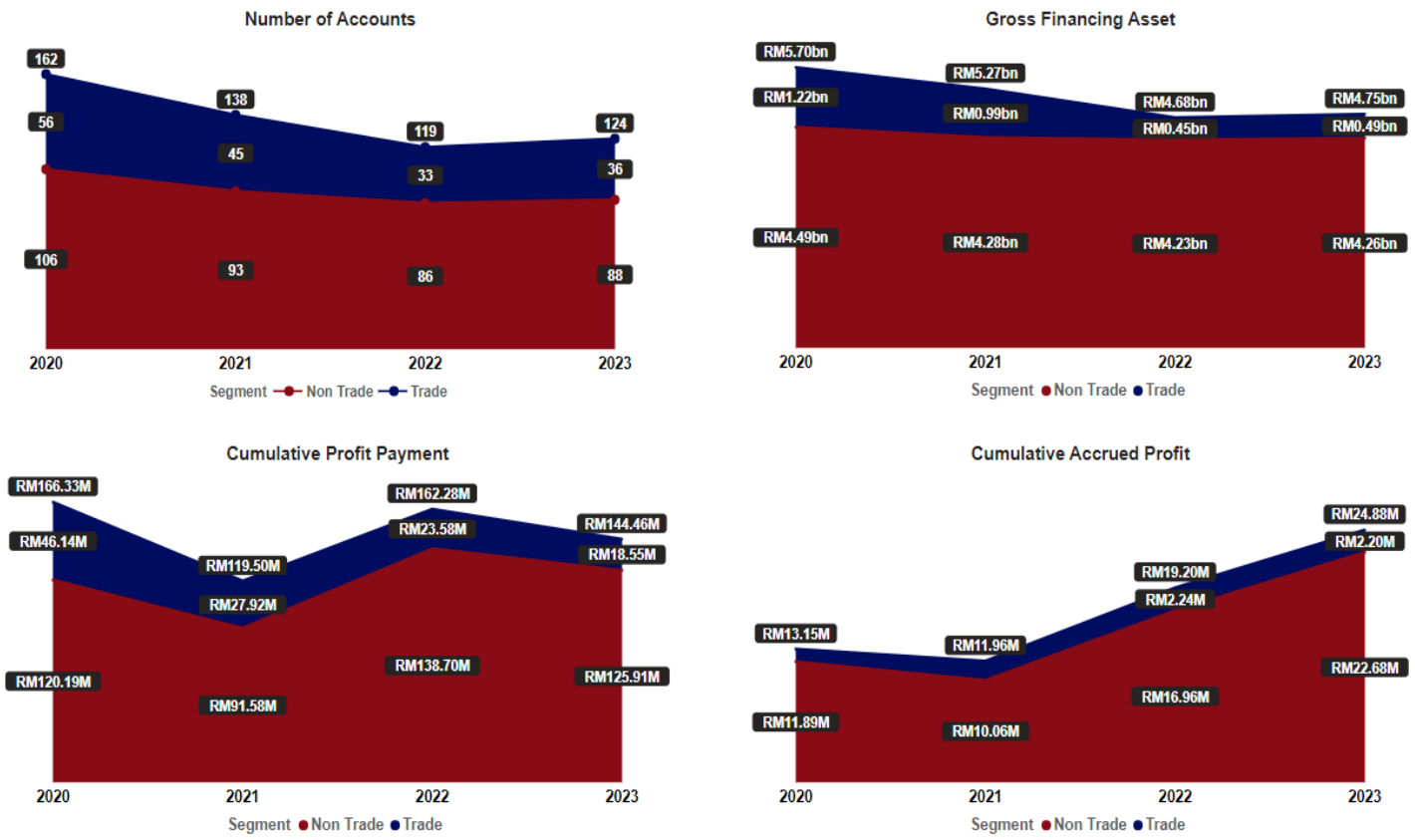


Figure 3.0

Figure above shows, the majority for number of accounts, gross financing asset, cumulative profit payment and cumulative accrued profit consist of non-trade compared to trade. Both trade and non-trade slowly decreases from year 2020 until year 2023 except for cumulative accrued profit that increases from year 2020 until year 2023.

Though most profit is from non-trade amounted between RM91M – RM140M while trade is between RM19M – RM47M, but in Figure 3.1 the profit security ratio will show Trade is more profitable whereby **higher ratio** means the profit is **more secure** and less asset is needed to earn profits or higher profit can be gain for trade if the cost for trade and non-trade is the same.

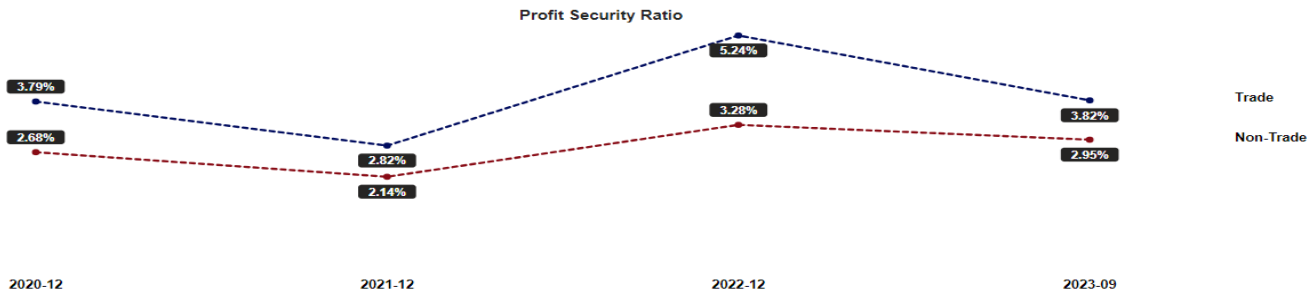


Figure 3.1

Although trade facility has an overall small portion for number of accounts, gross financing asset, cumulative profit payment and cumulative accrued profit. But the trade profit security ratio shows significantly higher that non-trade facility meaning a better security for the bank to generate profit and confident **trade is key to the bank 's long-term growth strategy**. Therefore, believing that our focus quality is paying off, trade financing will continue to grow in importance over time.

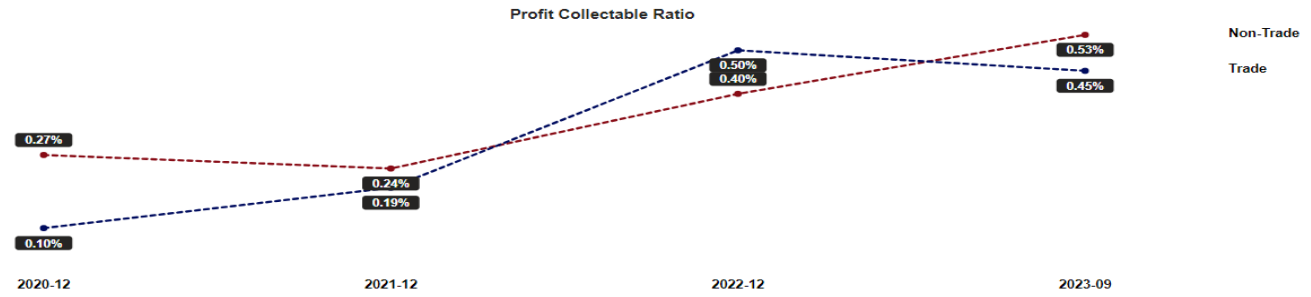


Figure 3.2

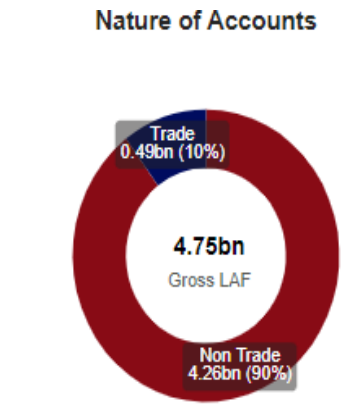
Even with the collectable ratio as per above, when combined with the security ratio the non-trade profits are still lower that trade profits.

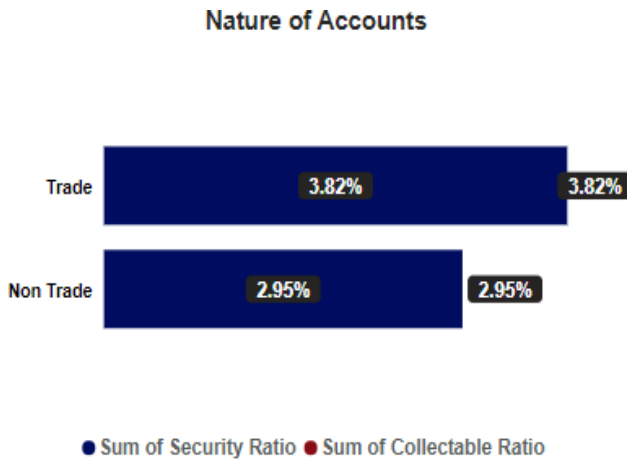
Security Portfolio

Figure 3.3 below will display EXIM bank **gross financing** and **overall profit ratio** and from EXIM clients as of the most recent position by 9 different criteria. The **most amounted financing asset does not directly indicate most profitable** whereby the higher the profit ratio the more secure the profit. List of criteria as per below:

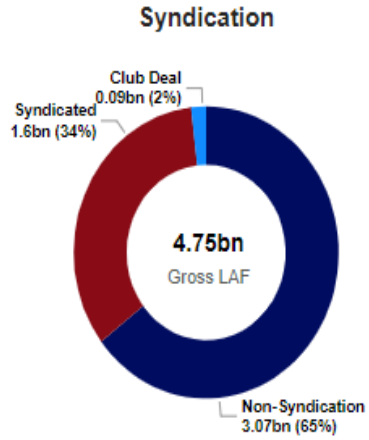
a) Nature of accounts
b) Syndication
c) Market type
d) Financing asset quality
e) Corporate Status
f) Entity type
g) Facility
h) Facility currency
i) Country
j) BNM main sector

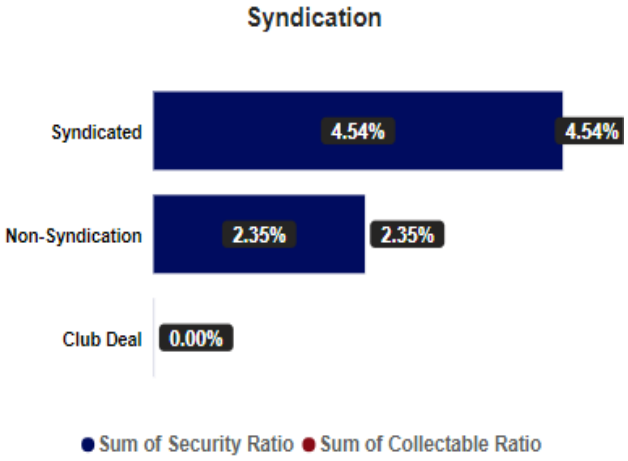
1- Nature of accounts



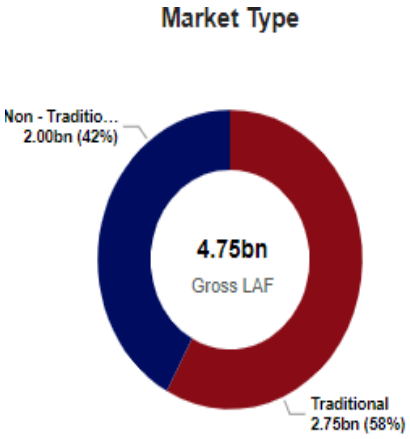


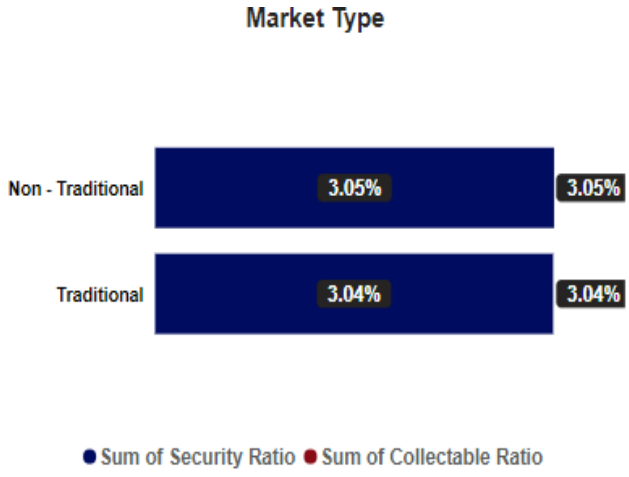
2- Syndication



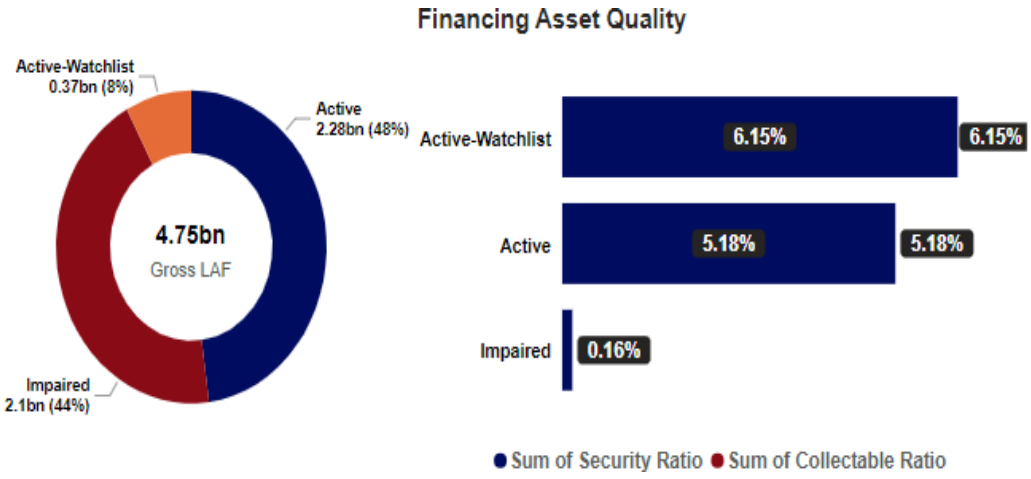


3- Market type

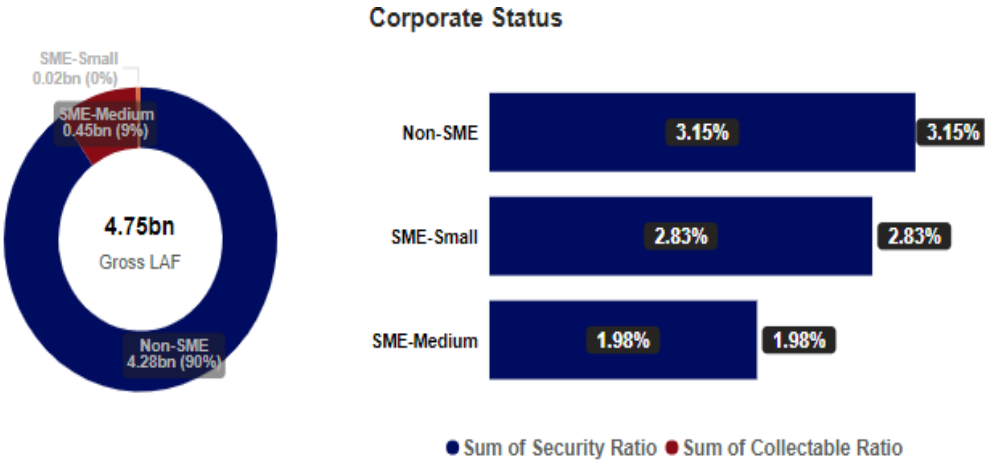




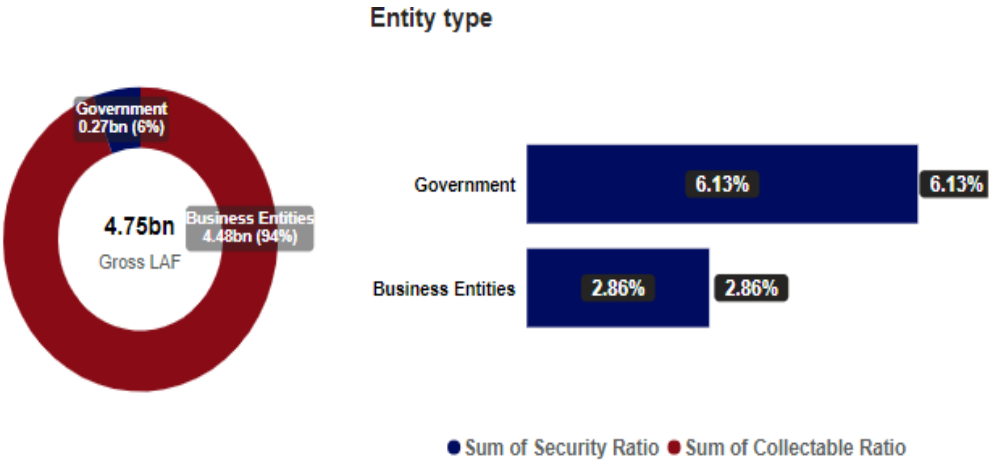
4- Financing asset quality



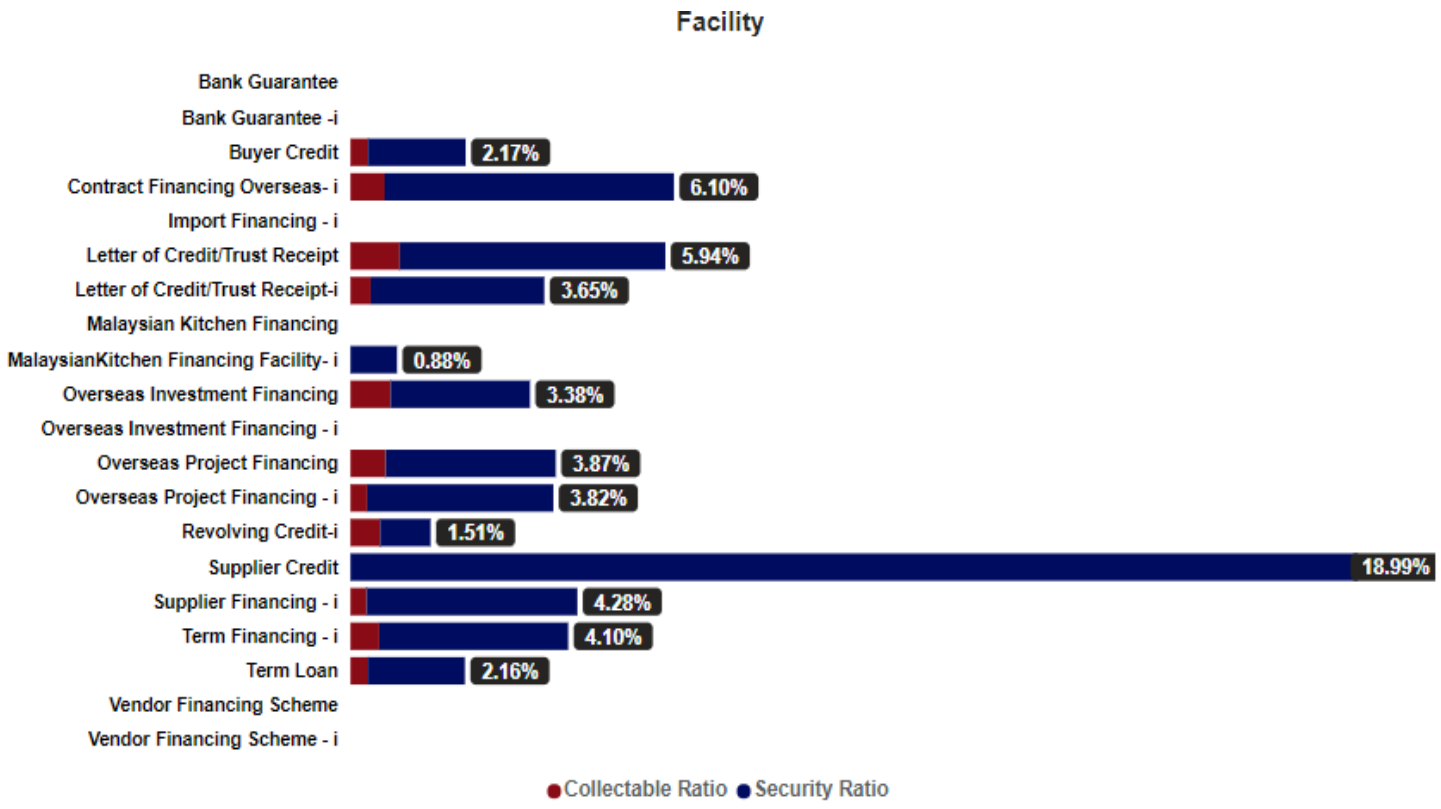
5- Corporate Status

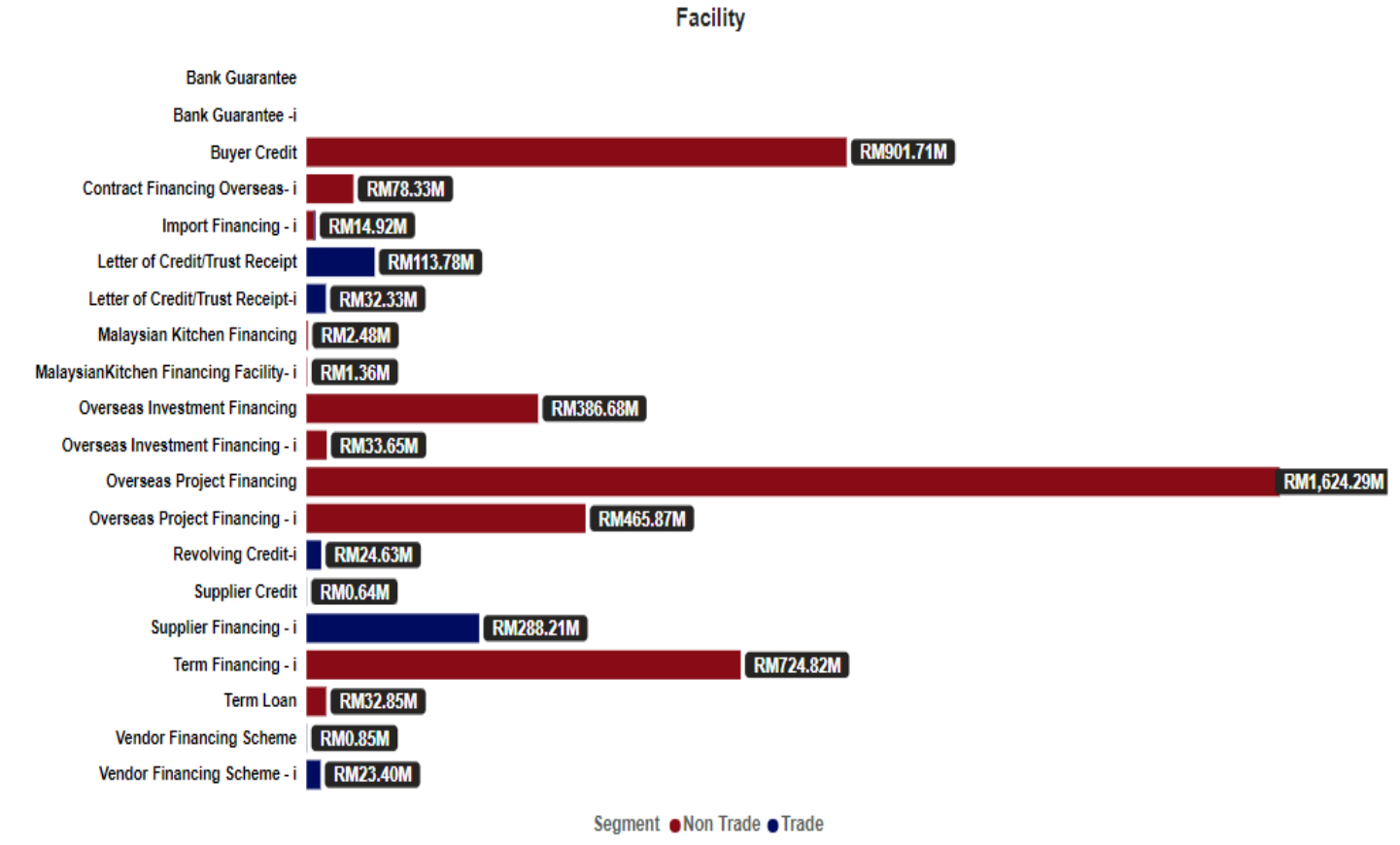


6- Entity type

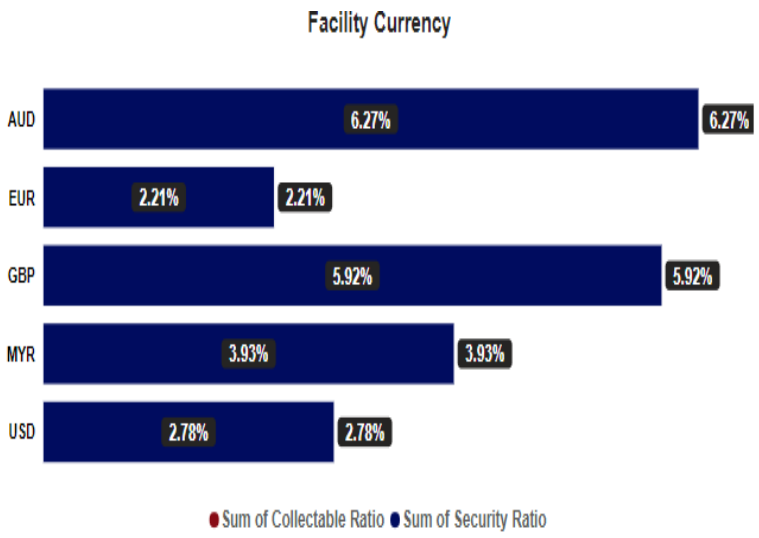


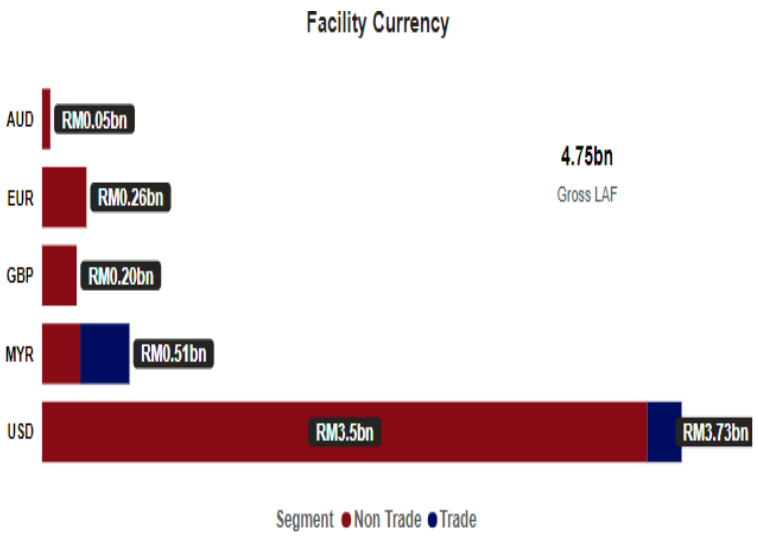
7- Facility


Facility
Bank Guarantee
Bank Guarantee -i
Buyer Credit
2.17%
Contract Financing Overseas- i
6.10%
Import Financing - i
Letter of Credit/Trust Receipt
5.94%
Letter of Credit/Trust Receipt-i
3.65%
Malaysian Kitchen Financing
MalaysianKitchen Financing Facility- i
0.88%
Overseas Investment Financing
3.38%
Overseas Investment Financing - i
Overseas Project Financing
3.87%
Overseas Project Financing - i
3.82%
Revolving Credit-i
1.51%
Supplier Credit
18.99%
Supplier Financing - i
4.28%
Term Financing - i
4.10%
Term Loan
2.16%
Vendor Financing Scheme
Vendor Financing Scheme - i
Collectable Ratio
Security Ratio

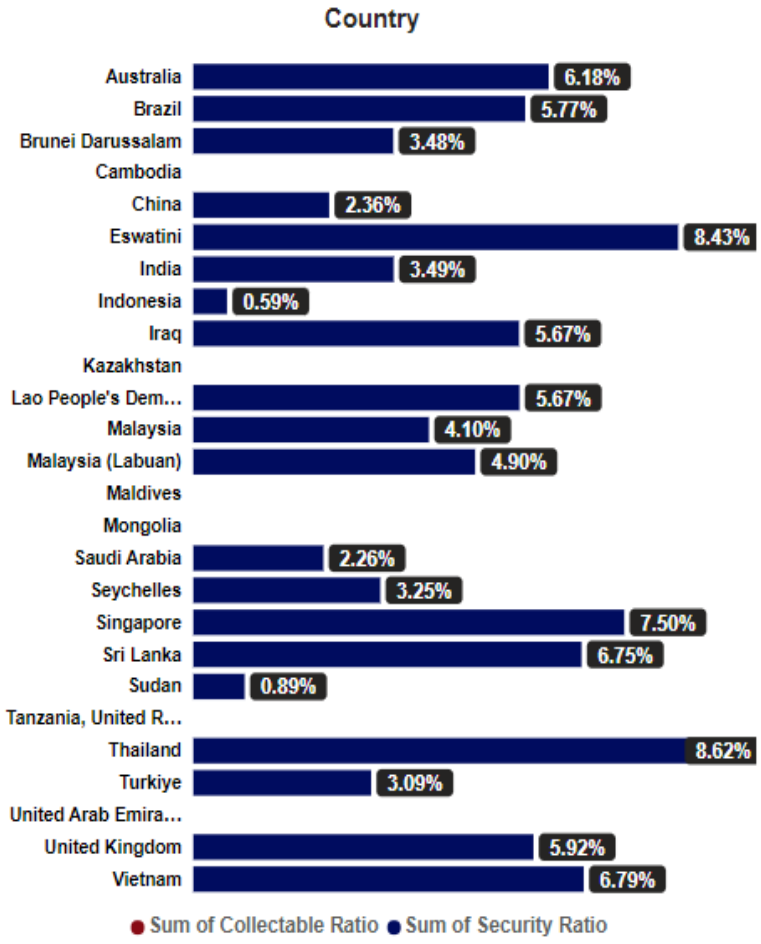

Facility
Bank Guarantee
Bank Guarantee -i
Buyer Credit
RM901.71M
Contract Financing Overseas- i
RM78.33M
Import Financing - i
RM14.92M
Letter of Credit/Trust Receipt
RM113.78M
Letter of Credit/Trust Receipt-i
RM32.33M
Malaysian Kitchen Financing
RM2.48M
MalaysianKitchen Financing Facility- i
RM1.36M
Overseas Investment Financing
RM386.68M
Overseas Investment Financing - i
RM33.65M
Overseas Project Financing
RM1,624.29M
Overseas Project Financing - i
RM465.87M
Revolving Credit-i
RM24.63M
Supplier Credit
RM0.64M
Supplier Financing - i
RM288.21M
Term Financing - i
RM724.82M
Term Loan
RM32.85M
Vendor Financing Scheme
RM0.85M
Vendor Financing Scheme - i
RM23.40M
Segment
Non Trade
Trade

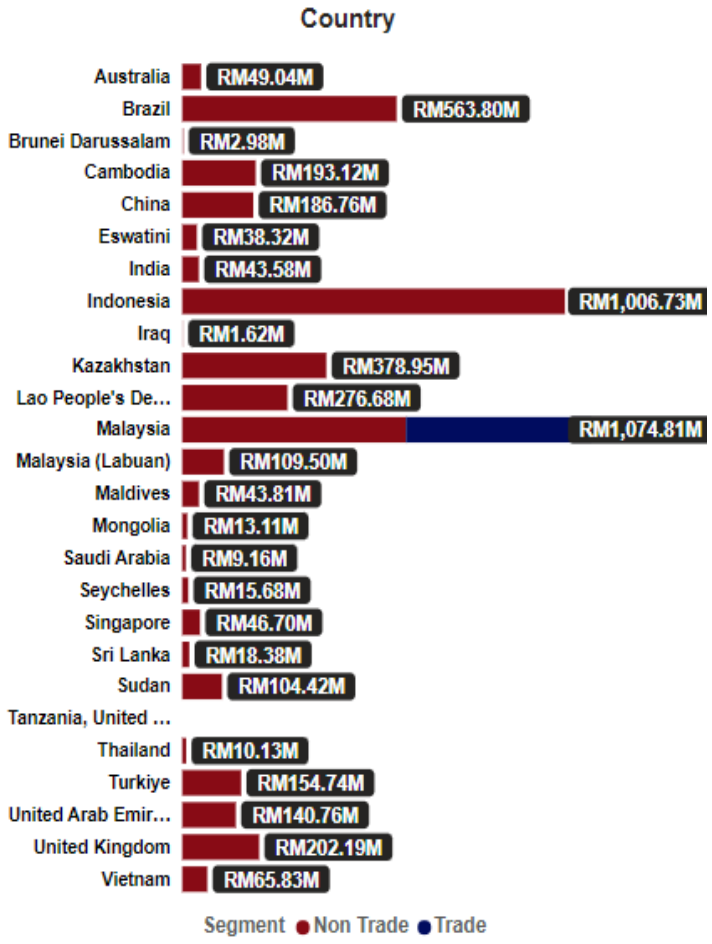
8- Facility currency





9- Country





10- BNM main sector

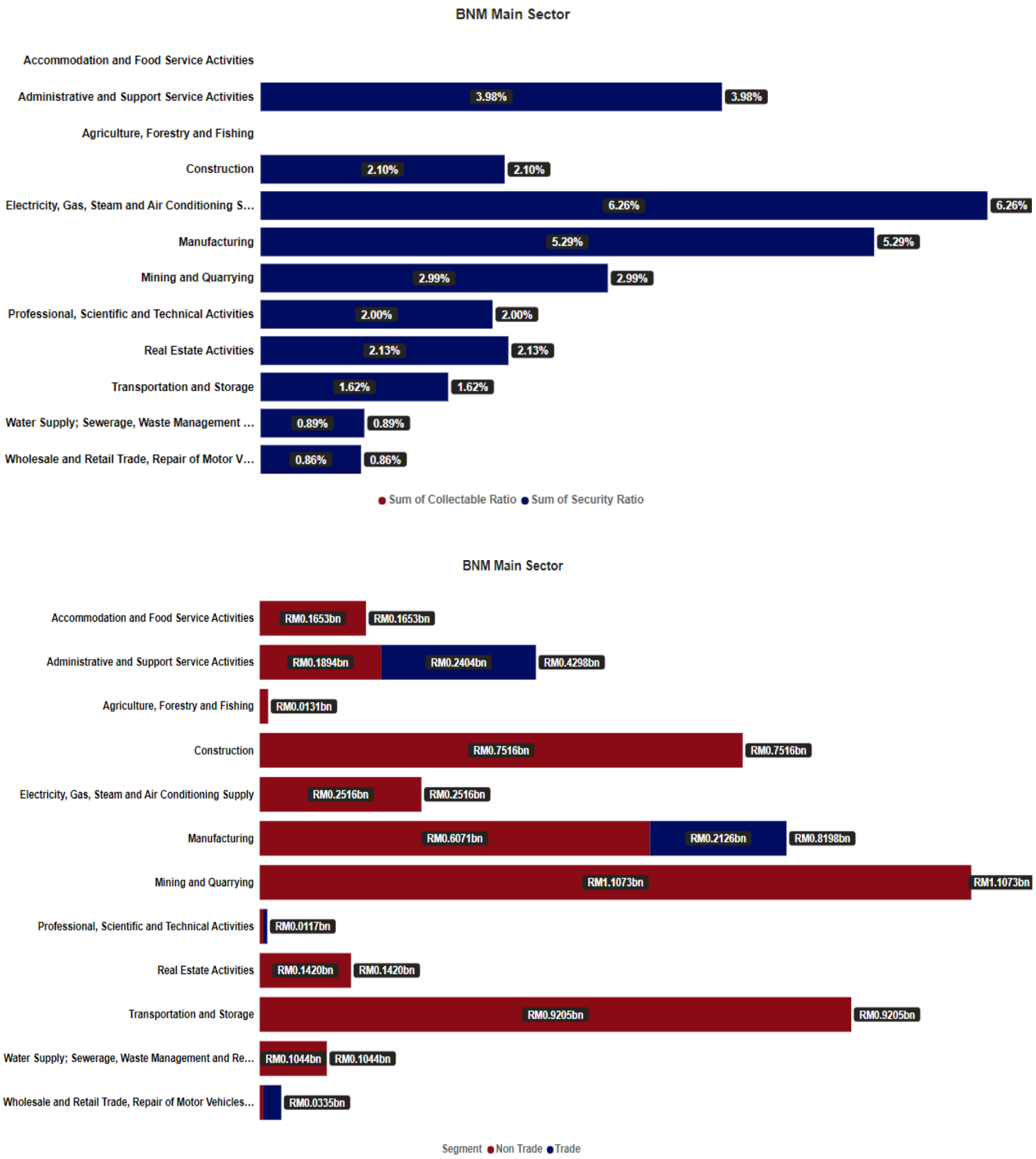


Figure 3.3

Security Criteria Prediction

- Trade secured profit score.

Trade financing has total of 24 branches/scenario, it predicted the capability of profit ranged from 0.023 bps to 0.420 bps security. Table below shows how are the most secured profit criteria:

Segment	Market Type	Payment Frequency	Corporate Status	Region	Country Rating	BNM Main Sector	Facility	Facility Currency	P-Value	Rank T/NT	Rank Overall
Trade	Traditional	Bullet	SME-Small	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	MYR	0.42	1	6
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	MYR	0.33	2	13
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	USD	0.32	3	17
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Supplier Financing - i	MYR	0.29	4	25
Trade	Traditional	Bullet	SME-Small	Asia	BBB+	Manufacturing	Supplier Credit	MYR	0.24	5	34
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt	USD	0.23	6	38
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Supplier Financing - i	MYR	0.21	7	44
Trade	Non - Traditional	Not Applicable	Non-SME	Africa	B	Manufacturing	Bank Guarantee	USD	0.21	8	45
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Letter of Credit/Trust Receipt-i	USD	0.21	9	46
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Supplier Financing - i	USD	0.21	10	47
Trade	Traditional	Not Applicable	SME-Medium	Asia	BB-	Manufacturing	Bank Guarantee	USD	0.19	11	50
Trade	Traditional	Bullet	SME-Small	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Import Financing - i	USD	0.18	12	53
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt	USD	0.16	13	55
Trade	Traditional	Not Applicable	SME-Medium	Asia	BBB+	Manufacturing	Supplier Financing - i	USD	0.15	14	56
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Vendor Financing Scheme	MYR	0.12	15	65
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Vendor Financing Scheme - i	MYR	0.11	16	66
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Manufacturing	Bank Guarantee	MYR	0.10	17	68
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Import Financing - i	USD	0.09	18	70
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Revolving Credit-i	USD	0.08	19	73
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Construction	Bank Guarantee	USD	0.07	20	75
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Administrative and Support Service Activities	Supplier Financing - i	MYR	0.07	21	76
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Bank Guarantee -i	USD	0.04	22	80
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Professional, Scientific and Technical Activities	Revolving Credit-i	MYR	0.03	23	81
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Real Estate Activities	Bank Guarantee	USD	0.02	24	82

Table 3.0

- Non-Trade secured profit score.

Non-Trade financing has total of 59 branches/scenario, it predicted the capibility of profit ranged from 0.021 bps to 0.662 bps security. Table below shows top 25 of the secured profit criteria:

Segment	Market Type	Payment Frequency	Corporate Status	Region	Country Rating	BNM Main Sector	Facility	Facility Currency	P-Value	Rank T/NT	Rank Overall
Non Trade	Non - Traditional	Quarterly	SME-Medium	Asia	CCC	Manufacturing	Overseas Project Financing - i	USD	0.66	1	1
Non Trade	Non - Traditional	Monthly	SME-Medium	Asia	Unrated	Mining and Quarrying	Term Financing - i	MYR	0.55	2	2
Non Trade	Non - Traditional	Monthly	Non-SME	Asia	CCC-	Construction	Overseas Project Financing	USD	0.49	3	3
Non Trade	Traditional	Monthly	SME-Small	Asia	BBB+	Manufacturing	Term Financing - i	MYR	0.46	4	4
Non Trade	Non - Traditional	Not Applicable	SME-Medium	Asia	B	Construction	Overseas Project Financing - i	USD	0.42	5	5
Non Trade	Non - Traditional	Monthly	SME-Small	Asia	Unrated	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	MalaysianKitchen Financing Facility- i	MYR	0.41	6	7
Non Trade	Traditional	Quarterly	SME-Small	Asia	BBB+	Construction	Overseas Project Financing	USD	0.40	7	8
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Manufacturing	Term Financing - i	MYR	0.36	8	9
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Manufacturing	Term Financing - i	USD	0.36	9	10
Non Trade	Non - Traditional	Monthly	SME-Medium	Asia	BBB-	Mining and Quarrying	Overseas Project Financing	USD	0.34	10	11
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	CCC+	Accommodation and Food Service Activities	Buyer Credit	USD	0.33	11	12
Non Trade	Non - Traditional	Semi annual	Non-SME	Middle East	CCC+	Construction	Term Loan	USD	0.32	12	14
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	CCC-	Electricity, Gas, Steam and Air Conditioning Supply	Buyer Credit	USD	0.32	13	15
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Transportation and Storage	Term Financing - i	MYR	0.32	14	16
Non Trade	Non - Traditional	Semi annual	Non-SME	Africa	BB-	Construction	Term Loan	USD	0.31	15	18
Non Trade	Non - Traditional	Monthly	SME-Small	Asia	BBB-	Accommodation and Food Service Activities	Malaysian Kitchen Financing	MYR	0.30	16	19
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB+	Manufacturing	Overseas Project Financing	USD	0.30	17	20
Non Trade	Non - Traditional	Monthly	SME-Medium	Africa	B-	Construction	Contract Financing Overseas- i	USD	0.30	18	21
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	BBB-	Mining and Quarrying	Overseas Project Financing	USD	0.29	19	22
Non Trade	Traditional	Monthly	Non-SME	Asia	BB-	Manufacturing	Overseas Project Financing - i	USD	0.29	20	23
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB	Manufacturing	Term Financing - i	MYR	0.29	21	24
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Term Financing - i	MYR	0.28	22	26
Non Trade	Non - Traditional	Semi annual	Non-SME	Europe	B	Transportation and Storage	Overseas Investment Financing	EUR	0.27	23	27
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB+	Transportation and Storage	Term Financing - i	USD	0.27	24	28
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Term Financing - i	USD	0.27	25	29

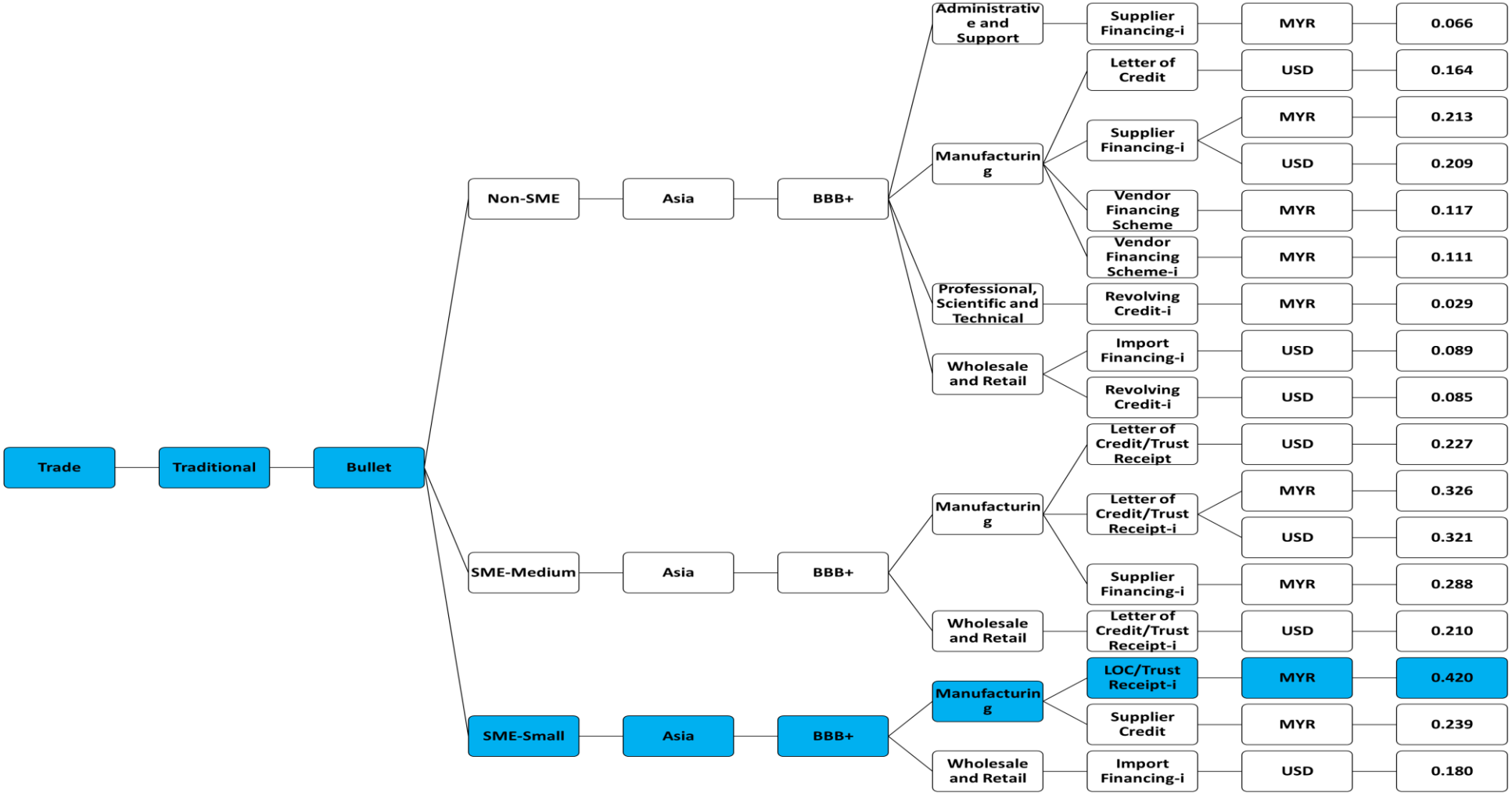
Table 3.1

- interpretation

From the graph/table resulted above, we could conclude the **biggest loan does not indicate the safest or highest profit**. Non-trade show highest ratio and the lowest ratio meaning though it can obtain most profit it also can end up with the least profit. Also, non-trade has more branches/scenario compared to trade indicates non-trade **has higher risk and higher sensitivity** for the profit and security to varies compared to trade.

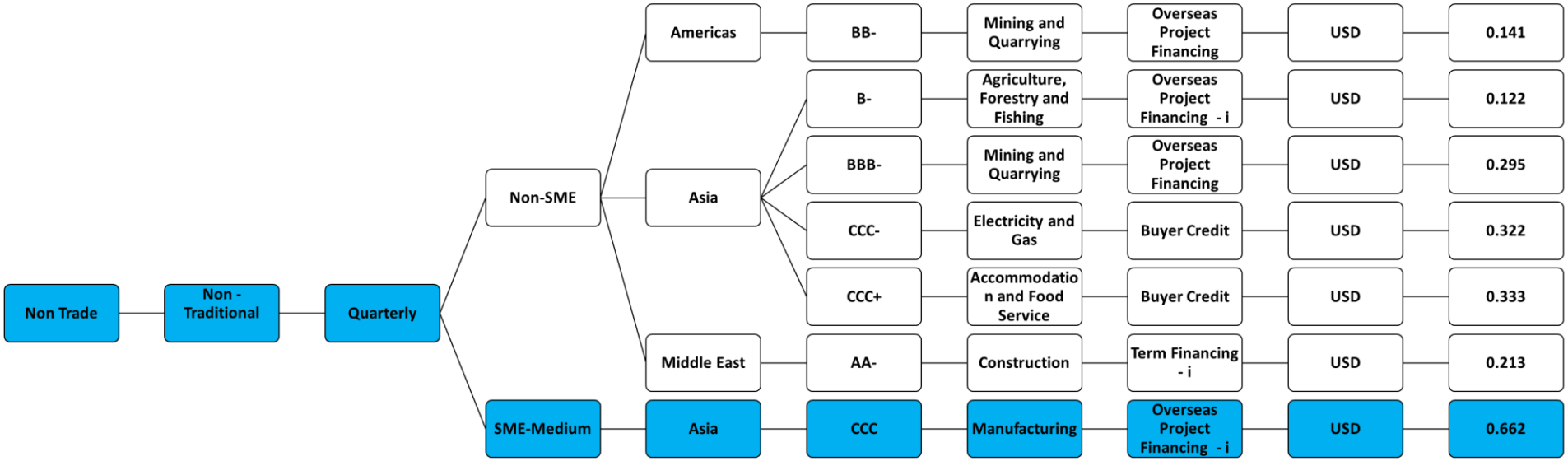
Profit Security Score (Trade)

Figure below shows sensitivity of 17 branches/scenario out of 24 branches link with the most profitable ratio:



Profit Security Score (Non-Trade)

Figure below shows sensitivity of 7 branches/scenario out of 59 branches link with the most profitable ratio:



CUSTOMER-CENTRIC: HOLISTIC APPROACH
FINANCING ASSET QUALITY
EXECUTIVE SUMMARY

Overall Financing Asset

Even while the **financing asset and cumulative profit payment is decreasing**, the active customer appears to be doing better as of September 2023 compared to December 2020. Though it is looking good for the active customer, at the time being we should constantly be concerned because the **impaired asset increases** from 36.41% to 44.16% where ECL will also increase:

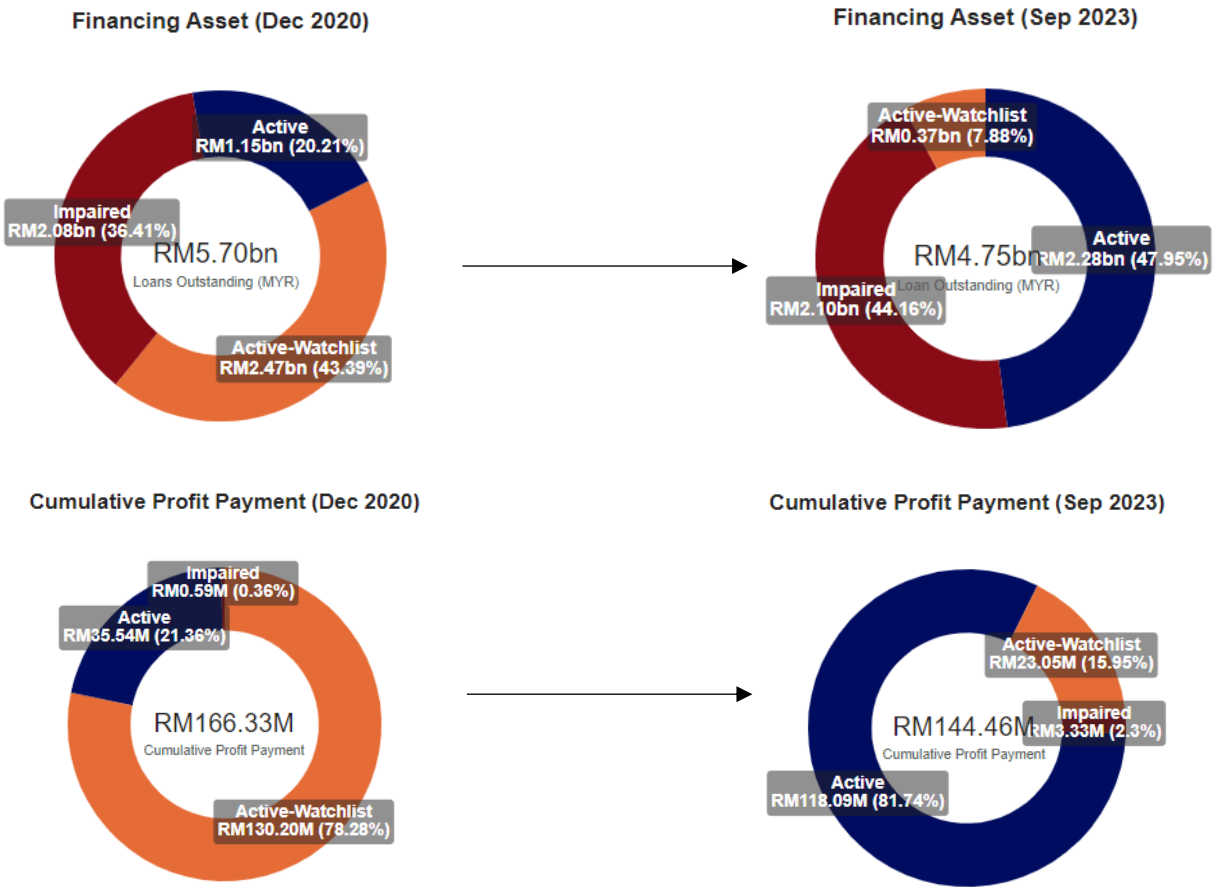


Figure 4.0

Since the decline of financing asset in the year 2020, it shows lesser watchlist customer means less ECL is allocated (1 year) which mean less cost. To decrease more cost, it advice to decrease the watchlist and impaired financing amount. Incontinuity, more cumulative profit can be generated because active customer makes more profit than watchlist and impaired customer:

There would be less cost and more quality profit for the bank when increase stage 1 (active) customer and decrease stage 2 (watchlist) and stage 3 (impaired) customer as per proven in the graph from December 2020 until September 2023 below:

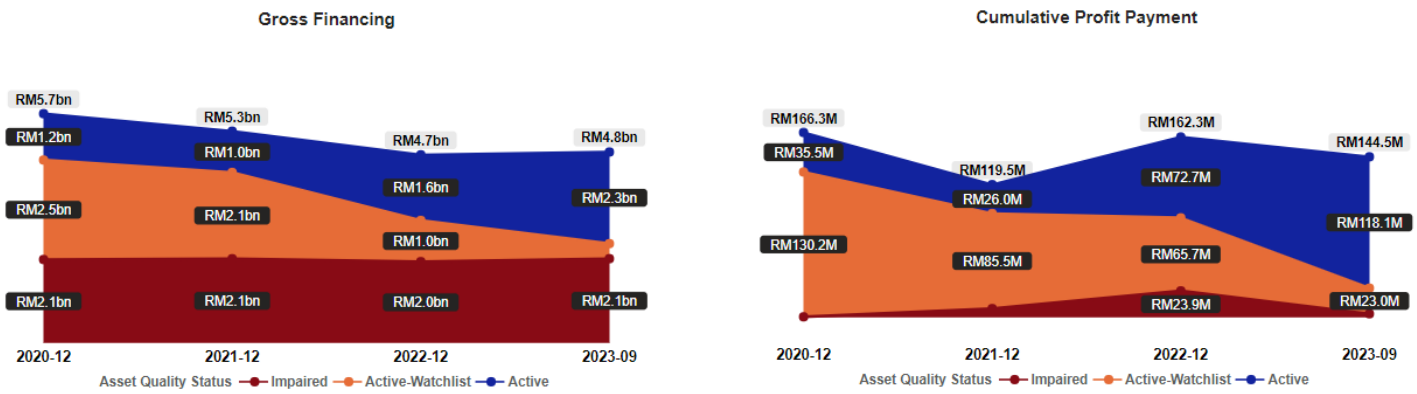


Figure 4.1

Figure below shows the number of impaired ratios as an overall, trade and non-trade also the number of impaired financings for trade and non-trade:

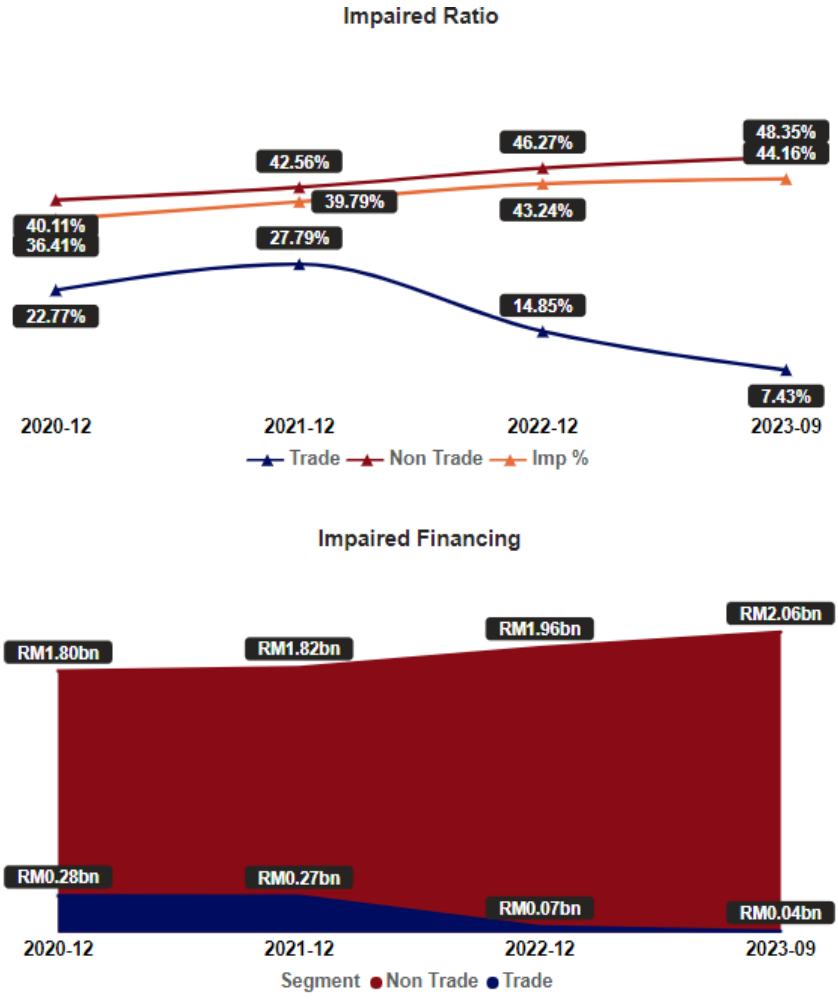


Figure 4.2

Overall impaired performance influenced by non-trade financing where the ratio shows the trade impaired ratio is decreasing from 22.77% to 7.43% while non-trade is increasing 40.11% to 48.35%. The total amount of impaired financing for non-trade increase from RM 1.80 billion to RM 2.06 billion while trade decrease from RM0.28 billion to RM 0.04 billion.

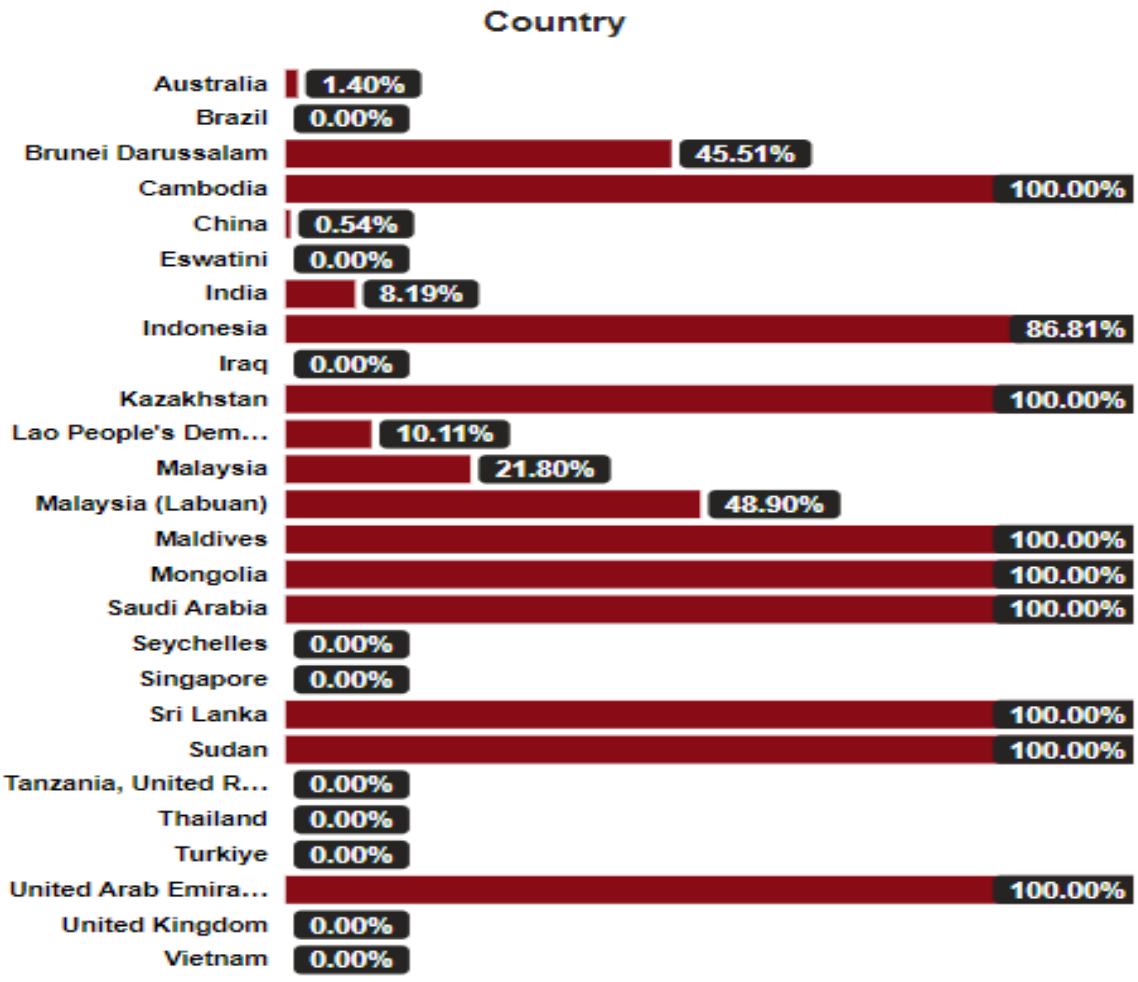
Impaired Ratio

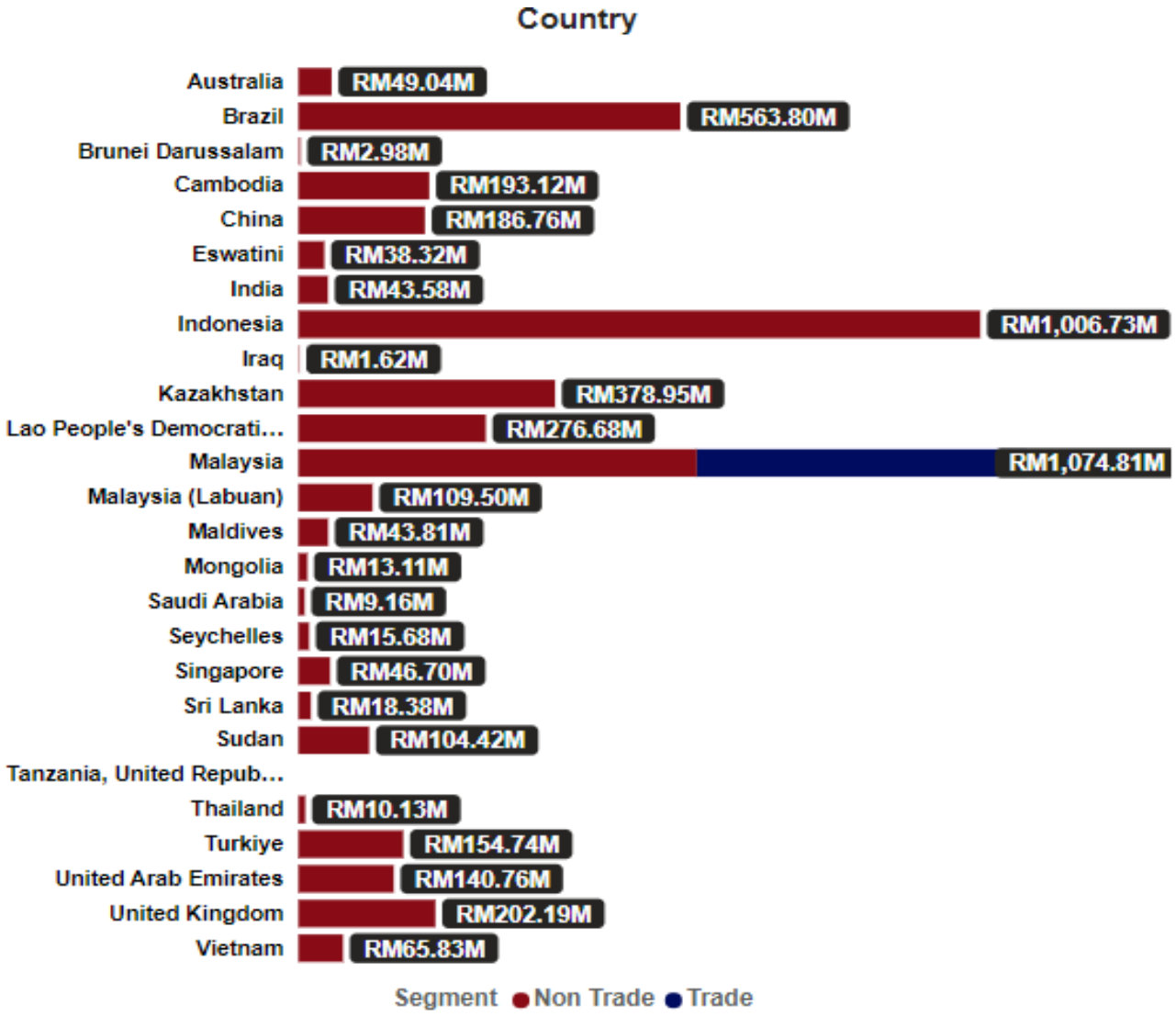
Figure 4.3 below will display **impaired ratio** and **gross financing** from EXIM clients as of the most recent position by 5 different criteria. The most amounted financing asset does not directly indicate highest profit whereby the higher the profit ratio the more secure the profit. List of criteria as per below:

a) Country
b) Currency
c) Facility
d) BNM Main Sector
e) Syndication

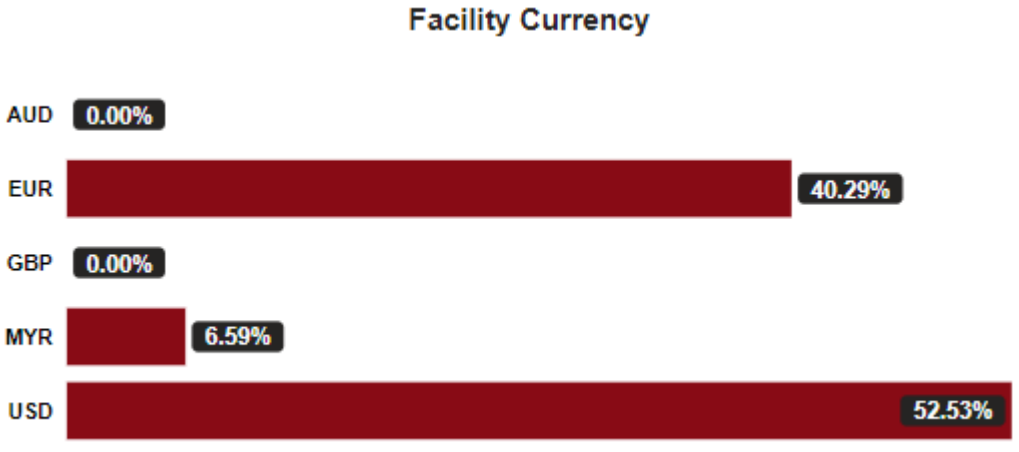
Impaired Ratio and Financing Asset:

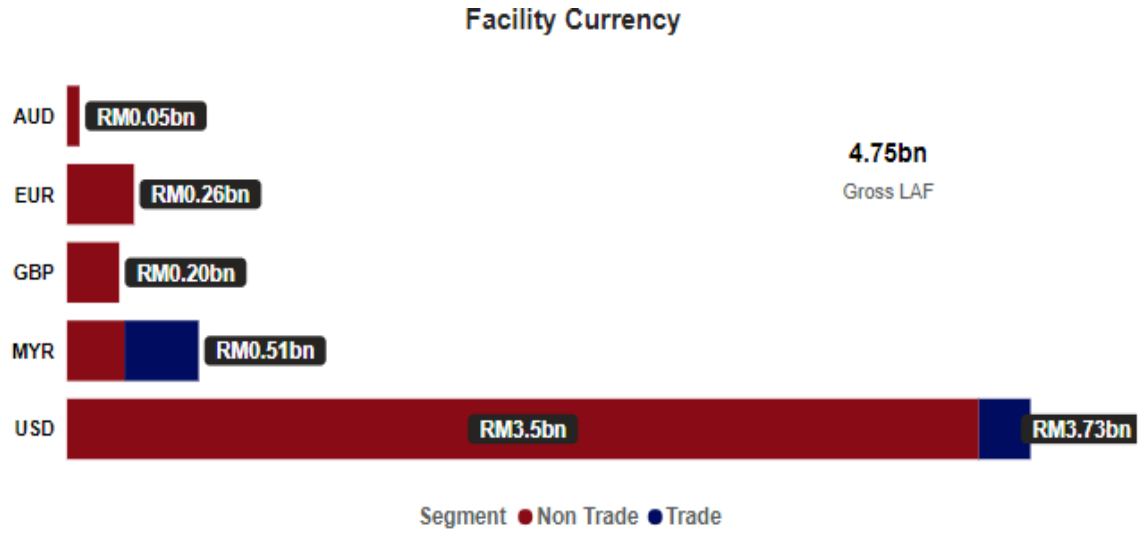
1- Country



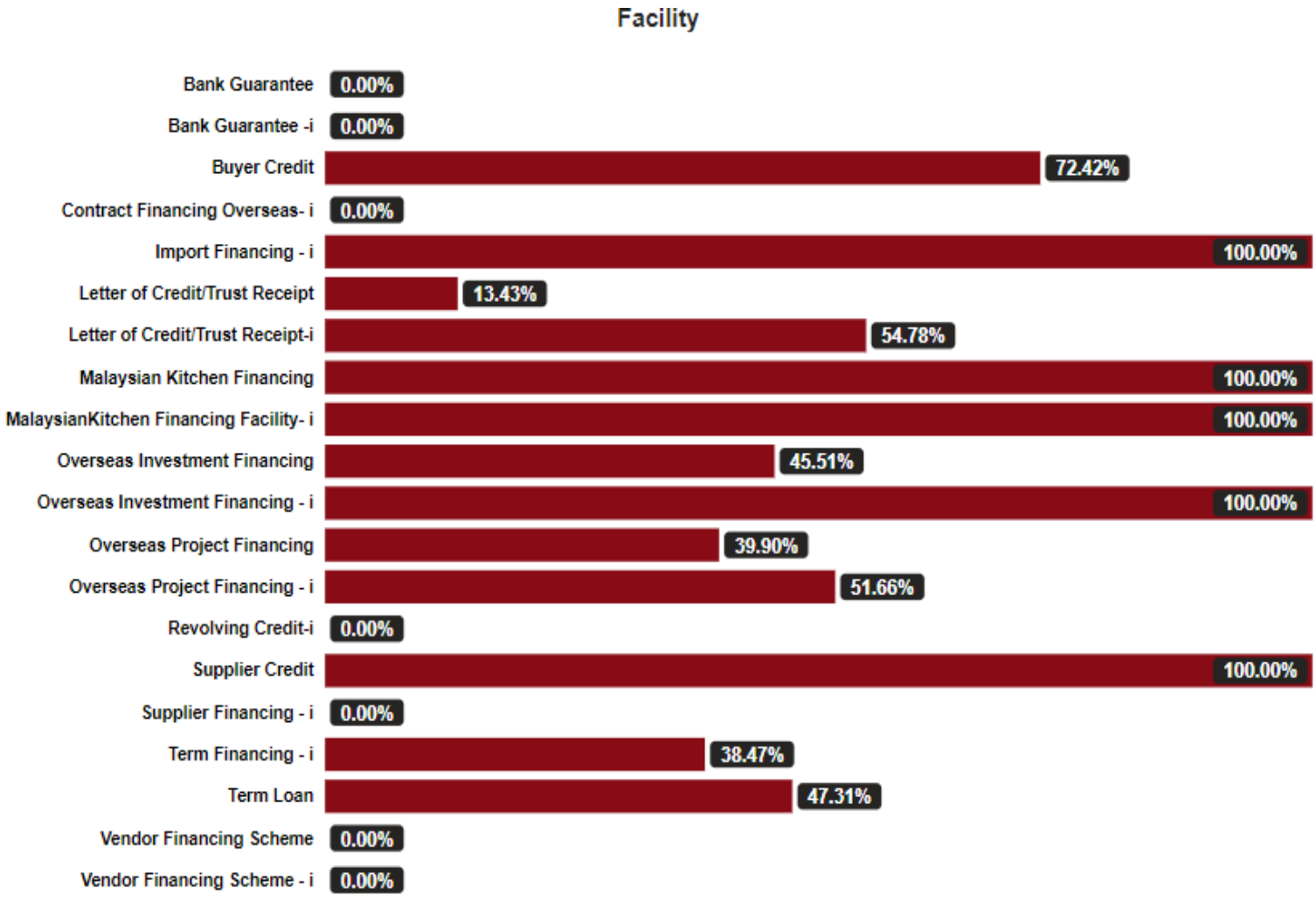

Country
Australia RM49.04M
Brazil RM563.80M
Brunei Darussalam RM2.98M
Cambodia RM193.12M
China RM186.76M
Eswatini RM38.32M
India RM43.58M
Indonesia RM1,006.73M
Iraq RM1.62M
Kazakhstan RM378.95M
Lao People's Democrati... RM276.68M
Malaysia RM1,074.81M
Malaysia (Labuan) RM109.50M
Maldives RM43.81M
Mongolia RM13.11M
Saudi Arabia RM9.16M
Seychelles RM15.68M
Singapore RM46.70M
Sri Lanka RM18.38M
Sudan RM104.42M
Tanzania, United Repub...
Thailand RM10.13M
Turkiye RM154.74M
United Arab Emirates RM140.76M
United Kingdom RM202.19M
Vietnam RM65.83M
Segment Non Trade Trade

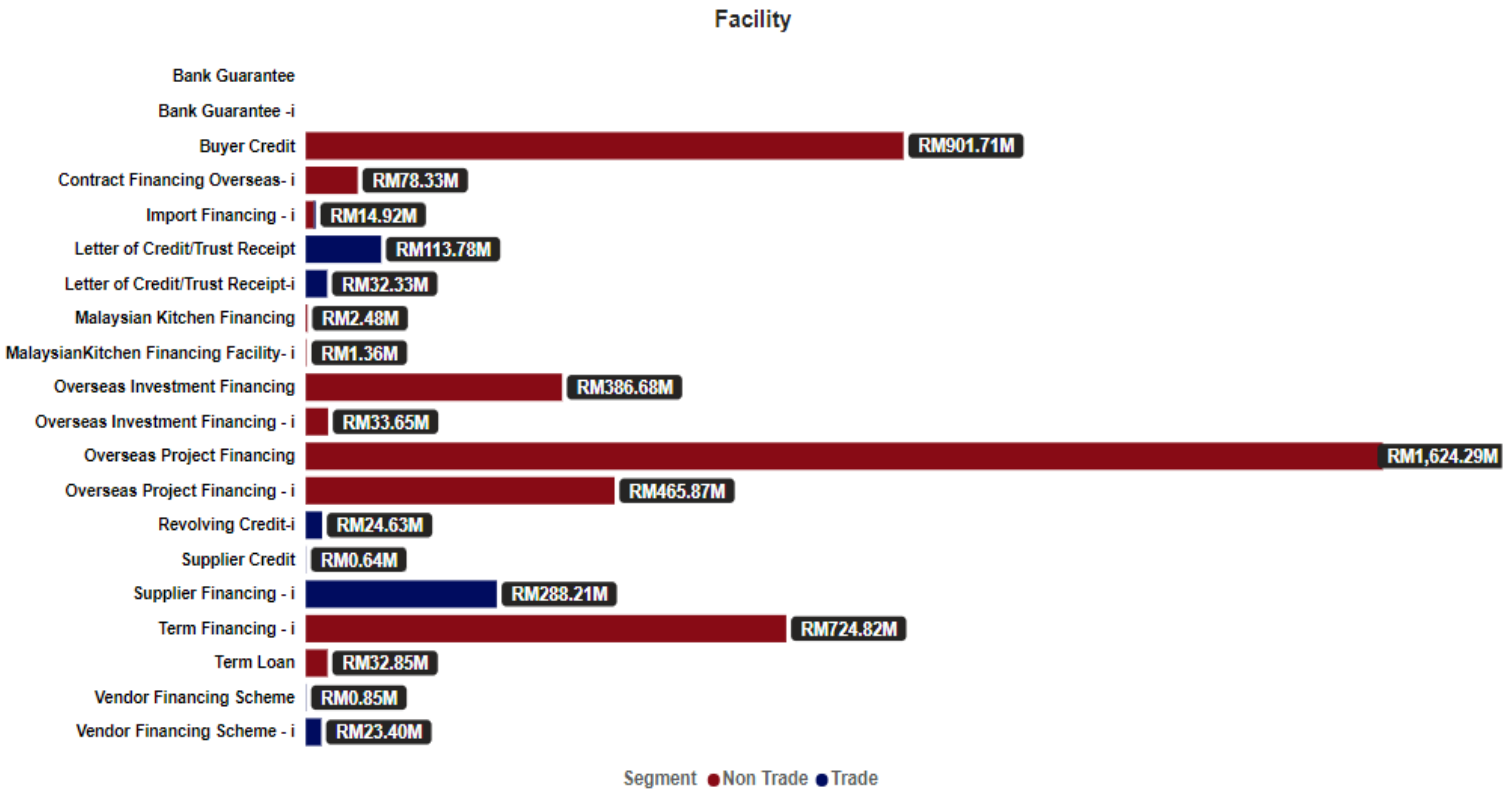
2- Currency


Facility Currency
AUD 0.00%
EUR 40.29%
GBP 0.00%
MYR 6.59%
USD 52.53%

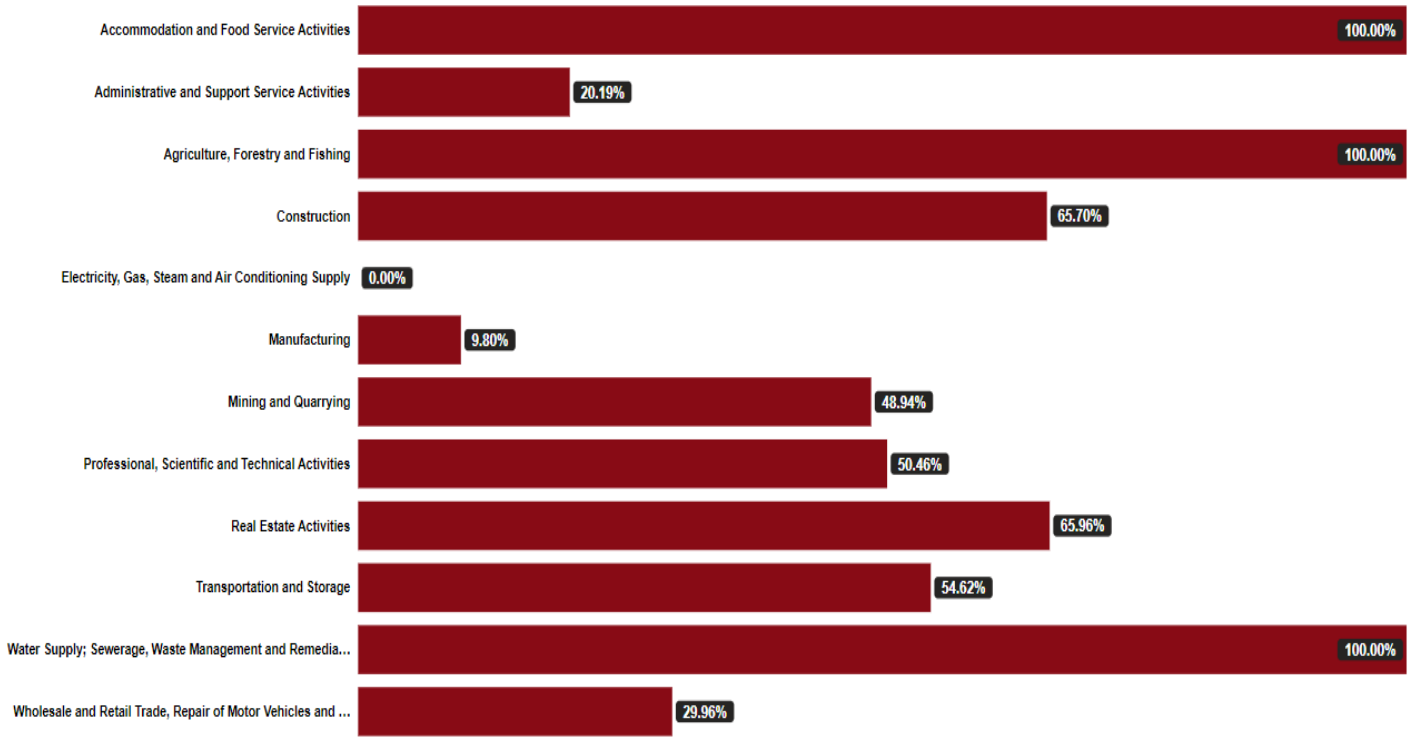

Facility Currency
AUD
RM0.05bn
EUR
RM0.26bn
GBP
RM0.20bn
MYR
RM0.51bn
USD
RM3.5bn
RM3.73bn
4.75bn
Gross LAF
Segment Non Trade Trade

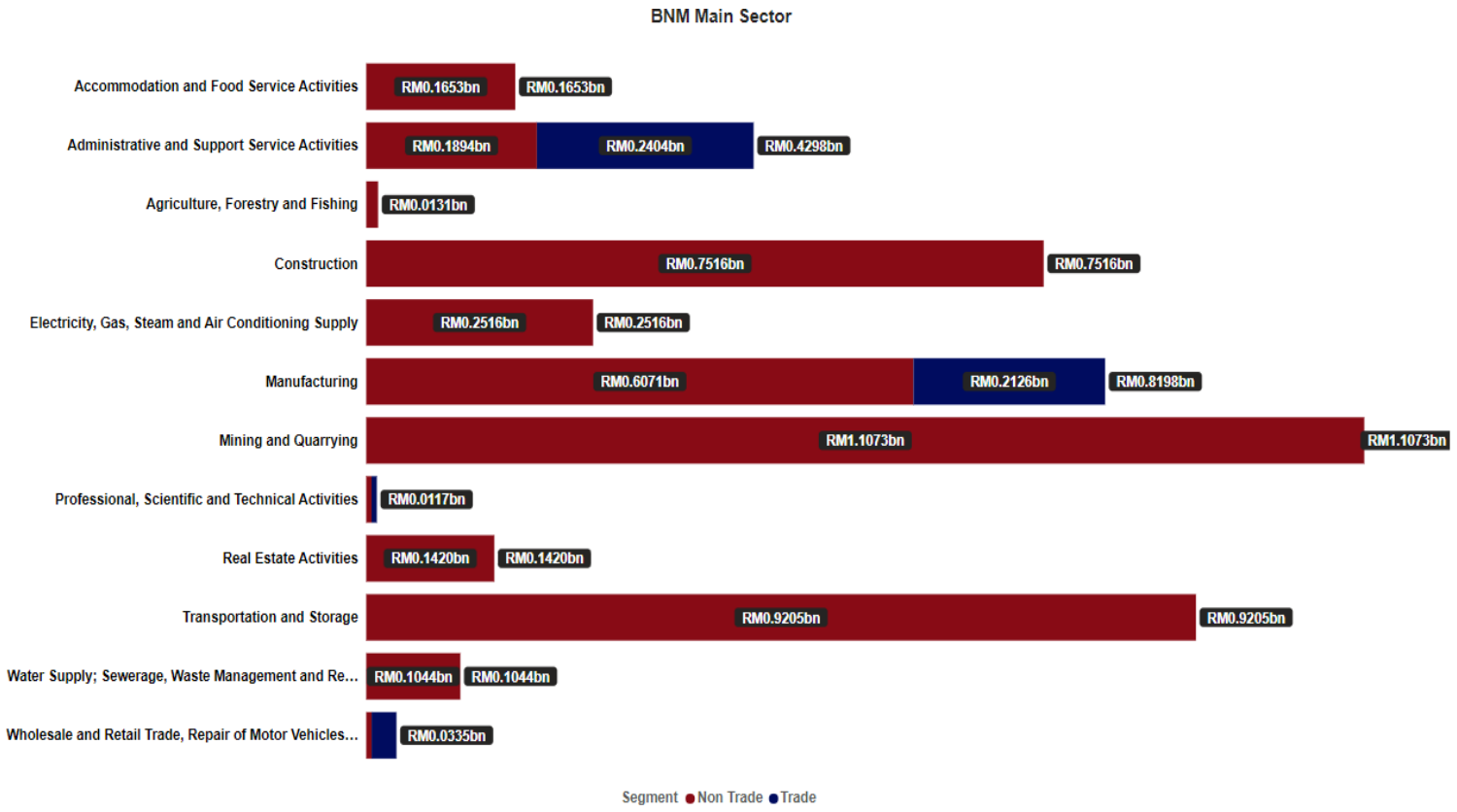
3- Facility


Facility
Bank Guarantee 0.00%
Bank Guarantee -i 0.00%
Buyer Credit 72.42%
Contract Financing Overseas- i 0.00%
Import Financing - i 100.00%
Letter of Credit/Trust Receipt 13.43%
Letter of Credit/Trust Receipt-i 54.78%
Malaysian Kitchen Financing 100.00%
MalaysianKitchen Financing Facility- i 100.00%
Overseas Investment Financing 45.51%
Overseas Investment Financing - i 100.00%
Overseas Project Financing 39.90%
Overseas Project Financing - i 51.66%
Revolving Credit-i 0.00%
Supplier Credit 100.00%
Supplier Financing - i 0.00%
Term Financing - i 38.47%
Term Loan 47.31%
Vendor Financing Scheme 0.00%
Vendor Financing Scheme - i 0.00%

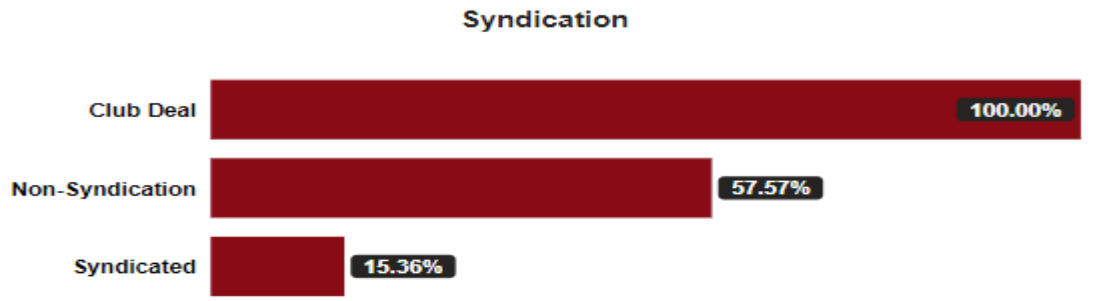


4- BNM Main Sector





5- Syndication



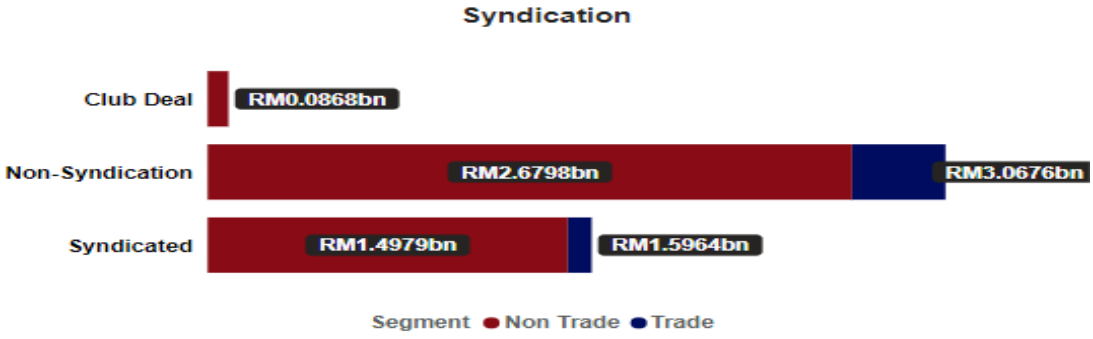


Figure 4.3

- interpretation

From the graph/table resulted above, we could conclude it proven the overall financing asset quality is improving influence by trade financing. **The bigger the loan does not directly have the best quality of customer** for example **Indonesia** number of financing asset but also has high impaired ratio. Also from the result, the most secure profit shows a **positive correlation** of good quality such as in country Thailand and Eswatini, even Sector in Electricity and Manufacturing activities.

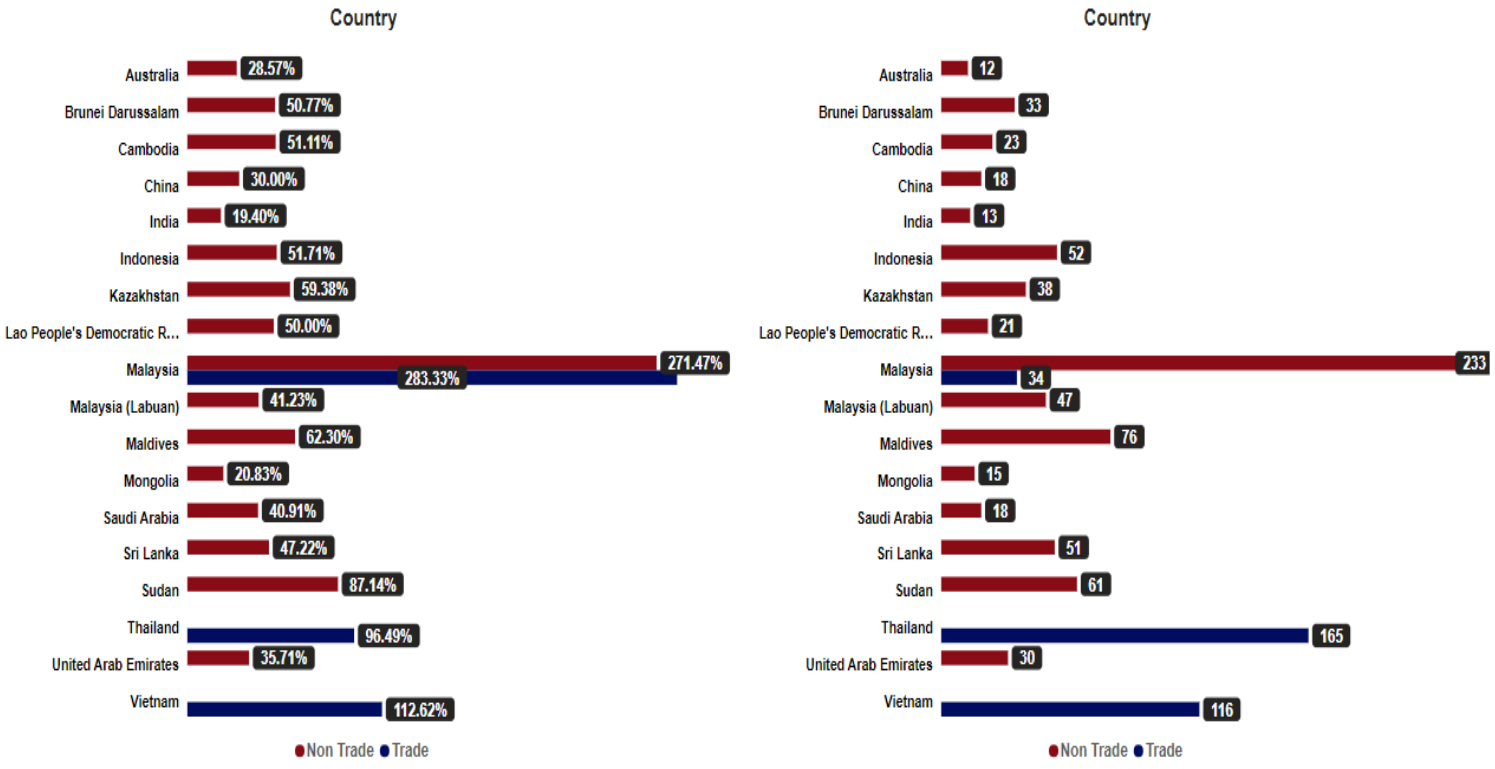
Impaired Behaviour

Figure 4.4 below will display **tenure to impaired ratio** and **number of months for certain criteria to turn impaired** as of the most recent position by 5 different criteria. List of criteria as per below:

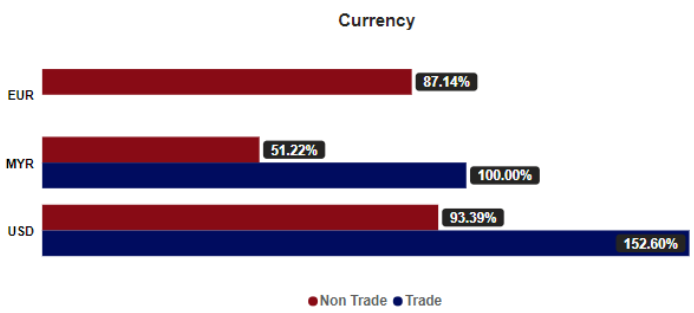
a) Country
b) Currency
c) Facility
d) BNM Main Sector
e) Syndication

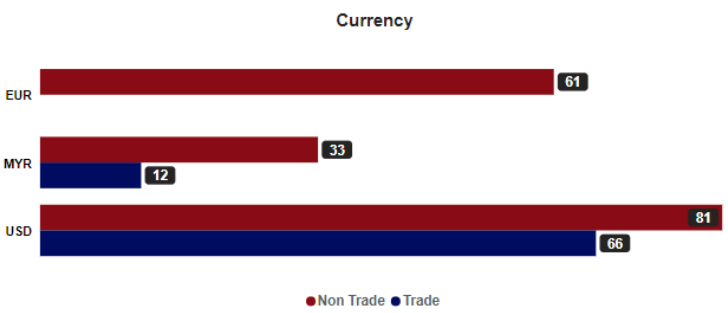
tenure to impaired ratio uses the average tenure and number of months for customer to turn impaired, a positive correlation will have a high positive ratio meaning **higher the ratio less likely for a criterion to turn impaired**, the 2nd graph will indicate how to turn impaired:

1- Country

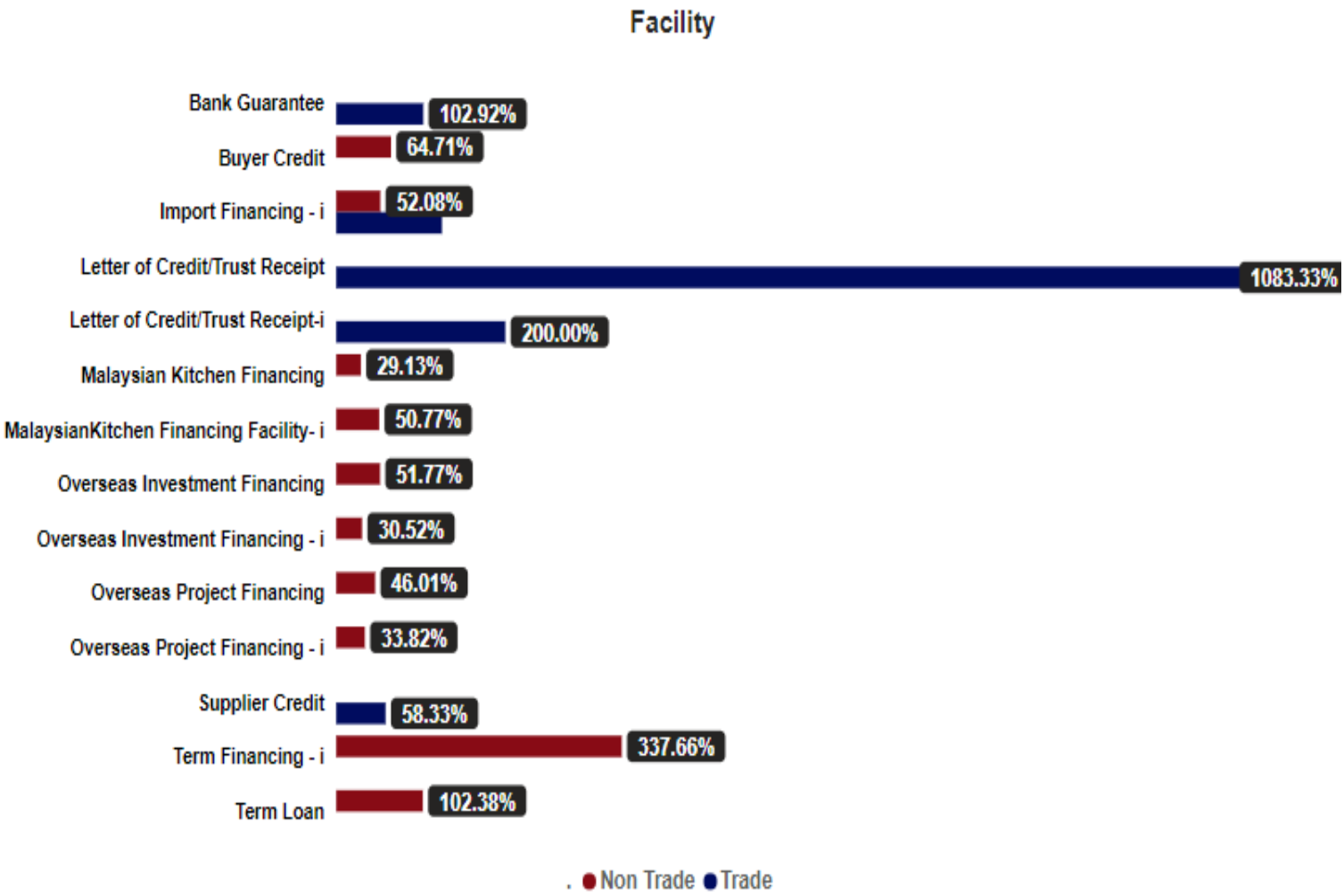


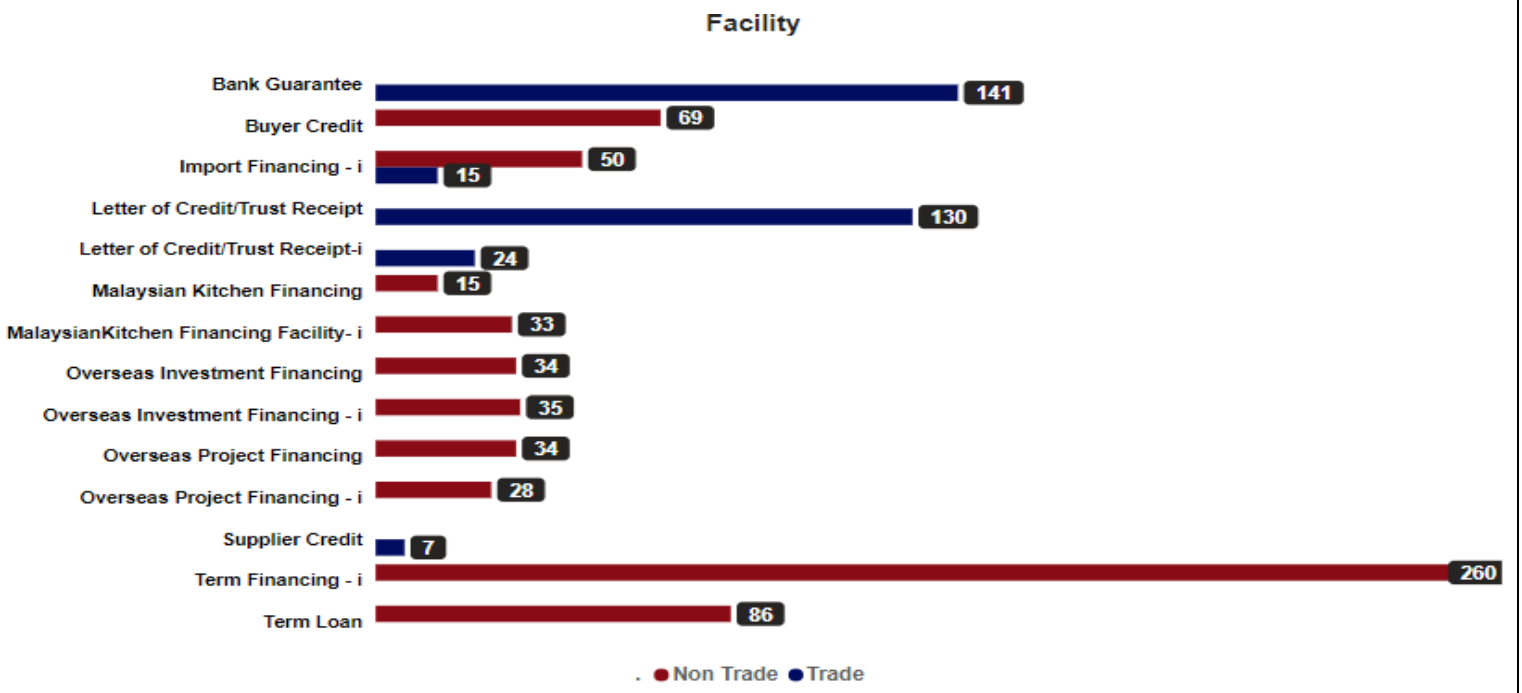
2- Currency



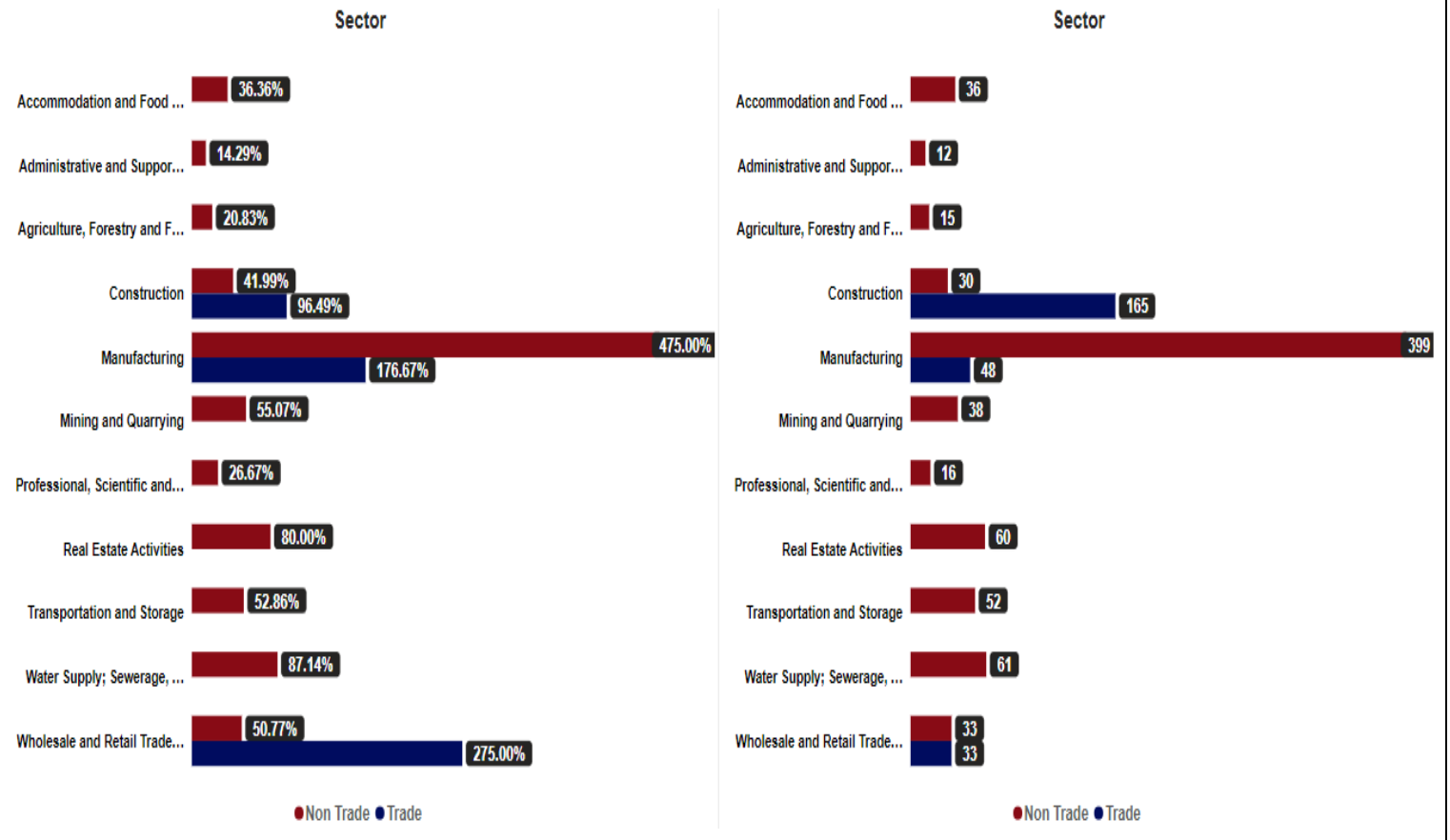


3- Facility





4- BNM Main Sector



5- Syndication

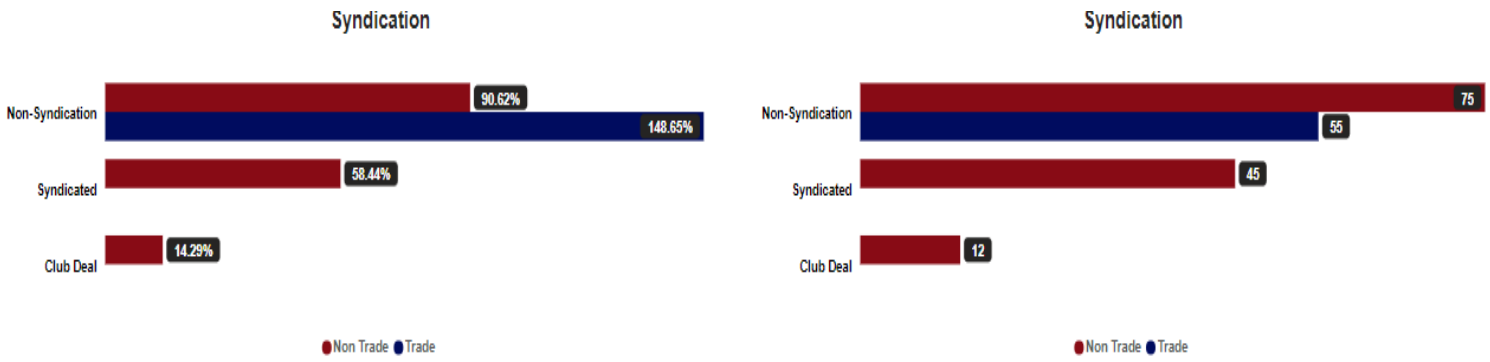


Figure 4.4

- interpretation

From the graph/table resulted above, most of non-trade criteria shows a possibility of how month most for a customer to turn impaired and trade has less likely to impaired. Example of Malaysia of trade and non-trade has the less most possibility to turn impaired while India and Mongolia have high chance of impaired by 13 months and 15 months respectively.

Accounts Migration

Table below shows number of accounts by asset quality segmentation throughout the year:

Financial Asset Quality	Dec 2020	Dec 2021	Dec 2022	Sep 2023
All Accounts	162	138	119	117
Active	37	17	26	46
Watchlist	55	49	31	19
Impaired	70	72	62	52

Table 4.0

Table below shows that the number of accounts, gross financing asset & cumulative profit payment for movement of an accounts. Overall, **new account increases slower than the churn accounts** by migration segmentation:

Overall Account				
Date	Dec 2020	Dec 2021	Dec 2022	Sep 2023
Total	162 Accounts	138 Accounts	119 Accounts	117 Accounts
Account Movement	-	24 Accounts	19 Accounts	2 Accounts
Gross Financing	RM5.70 billion	RM5.27 billion	RM4.68 billion	RM4.75 billion
C. Profit Payment	RM 0.17 billion	RM 0.12 billion	RM 0.16 billion	RM 0.14 billion

Maintain Accounts	-	129 Accounts	110 Accounts	101 Accounts
Gross Financing	-	RM4.81 billion	RM4.32 billion	RM4.49 billion
C. Profit Payment	-	RM0.12 billion	RM0.15 billion	RM0.14 billion

New Accounts	-	9 Accounts	9 Accounts	16 Accounts
Gross Financing	-	RM0.46 billion	RM0.37 billion	RM0.26 billion
C. Profit Payment	-	RM1.73 million	RM7.37 million	RM6.07 million

Churn Accounts	-	33 Accounts	28 Accounts	18 Accounts
Gross Financing	-	RM0.437 billion	RM0.802 billion	RM0.249 billion
C. Profit Payment	-	RM13.89 million	RM15.79 million	RM13.47 million
Reason	-	16 – Fully Settled 5 – Cancelled 5 – Conversion 4 – Expired 3 – Fully Write Off	14 – Fully Settled 9 – Fully Write Off 5 – Cancelled	8 – Fully Settled 8 – Fully Write-Off 1 – Cancelled 1 – Conversion
Cycle	-	1 new account disburse for dismissed accounts after December 2021 (Yinson Boronia Production B.V, Non-Trade, Overseas Project Financing, Settled Malaysia, New to Brazil)	No new accounts disburse for dismissed accounts after December 2022	-

Table 4.1

From the table below, the **number of impaired accounts is decreasing,** and **new impaired account is also decreasing** but most of the churn account in a **concern** when mostly the accounts in **fully written off** from EXIM book:

Impaired				
Date	Dec 2020	Dec 2021	Dec 2022	Sep 2023
Total	70 Accounts	72 Accounts	62 Accounts	52 Accounts
Account Movement	-	2 Accounts	10 Accounts	10 Accounts
Gross Financing	RM2.08 billion	RM2.10 billion	RM2.02 billion	RM2.10 billion
C. Profit Payment	RM0.59 million	RM7.99 million	RM23.92 million	RM3.33 million

Maintain Accounts	-	61 Accounts	55 Accounts	50 Accounts
Gross Financing	-	RM4.81 billion	RM1.79 billion	RM2.09 billion
C. Profit Payment	-	RM117.77 million	RM18.46 million	RM3.29 million

New Accounts	-	11 Accounts	7 Accounts	2 Accounts
Gross Financing	-	RM0.46 billion	RM0.23 billion	RM0.006 billion
C. Profit Payment	-	RM1.73 million	RM5.47 million	RM0.04 million

Churn Accounts	-	9 Accounts	17 Accounts	12 Accounts
Gross Financing	-	RM0.12 billion	RM0.37 billion	RM0.06 billion
C. Profit Payment	-	RM0.40 million	RM0.22 million	RM0.84 million

Reason	-	3 – Fully Write-Off 3 – Fully Settled 3 – Conversion	9 – Fully Write-Off 6 – Fully Settled 2 – Watchlist	8 – Fully Write-Off 3 – Watchlist 1 – Cancelled

Table 4.2

Cycle Diagram

Most of EXIM Bank customer is good churn when fully settled, so even the bad churn such as fully write-off is from impaired customer. Below shows the cycle diagrams for EXIM Bank customer:

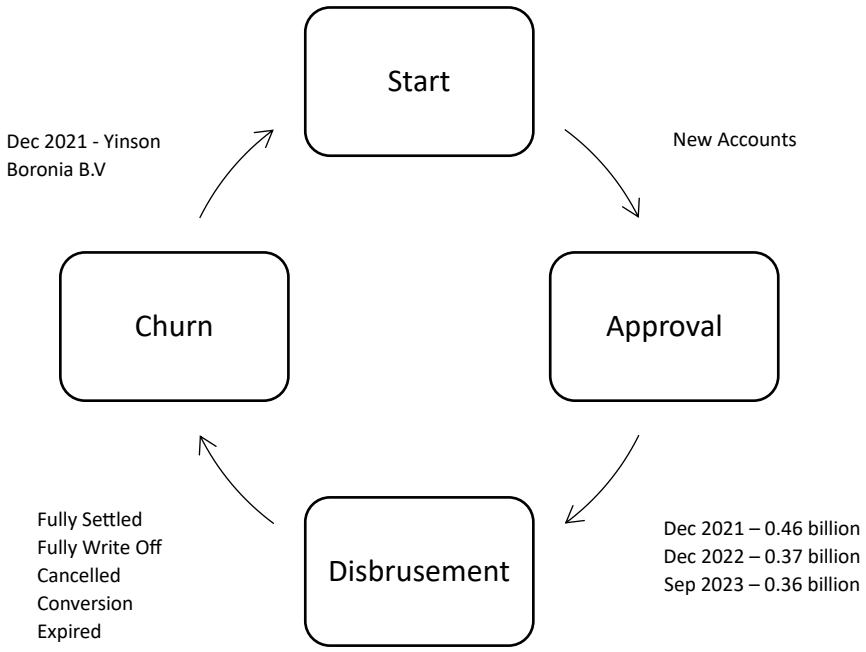


Diagram 4.0

Customer Churn Prediction

The prediction uses logistic regression model, one of supervised machine learning algorithm that is used to predict the probability of an outcome. The model can indicate whether an **active (46 Accounts)** will turn to **watchlist (19 Accounts)** and then whether it will turn **impaired (52 Accounts)** or not by setting up the correct parameter to predict those accounts. Listed parameter as per below:

1) Trade/Non-Trade Segmentation
2) Traditional/Non-Traditional Trade Market
3) Customer Payment Frequency
4) Customer Corporate Status
5) Country Region

6) S&P/Moodys Country Rating
7) Customer Main Sector
8) Customer Facility
9) Facility Currency

Matrix table:

Active to Watchlist	Suc-Obs	Fail_Obs
Suc-Pred	**11**	5
Fail-Pred	8	**41**
80.00%	57.89%	89.13%

Watchlist to Impaired	Suc-Obs	Fail_Obs
Suc-Pred	**50**	7
Fail-Pred	2	**12**
87.32%	96.15%	63.16%

Table 4.3

First matrix table: it predicted 8 (**RM118 million gross financing**) watchlist accounts will turn active by accuracy of 57.89% and 5 (**RM 168 million gross financing**) active accounts will turn watchlist by **89.13%.** The model shows accuracy of 80.00%.

Second matrix table: it predicted 2 (**RM 104 million gross financing**) impaired accounts will turn watchlist by accuracy of **96.15%** and 7 (**RM 330 million gross financing**) watchlist accounts will turn impaired by 63.16%. The model shows accuracy of 87.32%

Below shows the potential listing table based on the model:

1) Listing of possible active accounts to turn watchlist:

EXIM Account No.	Customer Name	1st Disbrusement	Country Exposure	Facility	Loans Outstanding (MYR)	p-Pred	Profit Security	Nature of Account	Asset Quality Status
3308-02138-202-0068-00	Bumi Armada Capital Offshore Ltd	2013	Malaysia (Labuan)	Overseas Investment Financing	55,955,241.11	0.73	9.59%	Non Trade	Active
3308-01137-107-0201-00	Lipidchem Sdn Bhd	2017	Malaysia	Term Financing - i	6,443,563.89	0.67	6.93%	Non Trade	Active
3308-02218-104-0254-00	Probase Eswatini Pty Ltd	2020	Eswatini	Contract Financing Overseas- i	38,324,268.86	0.90	8.43%	Non Trade	Active
3308-01137-107-0296-02	UZMA KUALA MUDA SDN. BHD	2023	Malaysia	Term Financing - i	1,483,059.00	0.52	0.00%	Non Trade	Active
3308-02137-101-0270-00	Whitex Garments Sdn Bhd	2021	Vietnam	Overseas Project Financing - i	65,833,593.39	0.54	6.79%	Non Trade	Active
Total					168,039,726.25		31.75%		

2) Listing of possible watchlist accounts to turn active:

EXIM Account No.	Customer Name	1st Disbrusement	Country Exposure	Facility	Loans Outstanding (MYR)	p-Pred	Profit Security	Nature of Account	Asset Quality Status
3308-02137-109-0087-00	Destini Prima Sdn Bhd	2014	Malaysia	Import Financing - i	0.00	0.09	0.00%	Trade	Active-Watchlist
3308-02137-101-0246-00	Cahya Mata Phosphates Industries Sdn Bhd (formerly known as Malaysian Phosphate Additives (Sarawak) Sdn Bhd	2019	Malaysia	Overseas Project Financing - i	64,332,803.02	0.18	6.36%	Non Trade	Active-Watchlist
3308-02203-200-0020-00	Republic of Seychelles	2009	Seychelles	Term Loan	15,680,516.57	0.25	3.25%	Non Trade	Active-Watchlist
3308-02137-117-0071-00	Destini Prima Sdn Bhd	2014	Malaysia	Bank Guarantee -i	0.00	0.32	0.00%	Trade	Active-Watchlist
3308-01137-107-0260-00	SSP Raw Metals Trading Sdn Bhd	2020	Malaysia	Term Financing - i	1,687,375.04	0.35	6.88%	Non Trade	Active-Watchlist
3308-01137-107-0209-01	PWN Excellence Sdn Bhd	2017	Malaysia	Term Financing - i	16,533,989.48	0.43	7.06%	Non Trade	Active-Watchlist
3308-01137-107-0209-03	PWN Excellence Sdn Bhd	2017	Malaysia	Term Financing - i	18,308,667.03	0.43	5.35%	Non Trade	Active-Watchlist
3308-02107-200-0008-00	Republic of Iraq	2008	Iraq	Term Loan	1,624,530.30	0.44	5.67%	Non Trade	Active-Watchlist
Total					118,167,881.45		34.56%		

3) Listing of possible watchlist accounts to turn impaired:

EXIM Account No.	Customer Name	1st Disbrusement	Country Exposure	Facility	Total Loans Outstanding (MYR)	p-Pred	Profit Security	Nature of Account	Asset Quality Status
3308-02203-200-0020-00	Republic of Seychelles	2009	Seychelles	Term Loan	15,680,516.57	0.96	3.25%	Non Trade	Active-Watchlist
3308-02137-109-0087-00	Destini Prima Sdn Bhd	2014	Malaysia	Import Financing - i	0.00	0.94	0.00%	Trade	Active-Watchlist
3308-02107-200-0008-00	Republic of Iraq	2008	Iraq	Term Loan	1,624,530.30	0.91	5.67%	Non Trade	Active-Watchlist
3308-02137-101-0246-00	Cahya Mata Phosphates Industries Sdn Bhd (formerly known as Malaysian Phosphate Additives (Sarawak) Sdn Bhd	2019	Malaysia	Overseas Project Financing - i	64,332,803.02	0.85	6.36%	Non Trade	Active-Watchlist
3308-02137-117-0071-00	Destini Prima Sdn Bhd	2014	Malaysia	Bank Guarantee -i	0.00	0.75	0.00%	Trade	Active-Watchlist
3308-02123-205-0022-00	The Ministry of Finance Government of Lao PDR	2010	Lao People's Democratic Republic	Buyer Credit	7,631,769.82	0.73	17.69%	Non Trade	Active-Watchlist
3308-02123-205-0204-00	The Ministry of Finance Government of Lao PDR	2017	Lao People's Democratic Republic	Buyer Credit	241,063,762.79	0.73	5.95%	Non Trade	Active-Watchlist
Total					330,333,382.50		38.92%		

4) Listing of possible impaired accounts to turn watchlist:

EXIM Account No.	Customer Name	1st Disbrusement	Country Exposure	Facility	Total Loans Outstanding (MYR)	p-Pred	Profit Security	Nature of Account	Asset Quality Status
3308-02137-216-0048-00	Mir Valve Sdn Bhd	2013	Vietnam	Bank Guarantee	0.00	0.38	0.00%	Trade	Impaired
3308-04215-205-0027-00	Al Manara Water Company Limited	2011	Sudan	Buyer Credit	104,422,553.53	0.11	0.89%	Non Trade	Impaired
Total					104,422,553.53		0.89%		

The Receiver Operating Characteristic (ROC) curves illustrates the diagnostic ability to determine for an account to churn or vice versa. Under the ROC curves the overall account performance based on the 9 parameters listed above, if the probability of impairment for an account is above 0.5 in figure below, we consider for the account as churn while below 0.5 says otherwise.

ROC Curve:

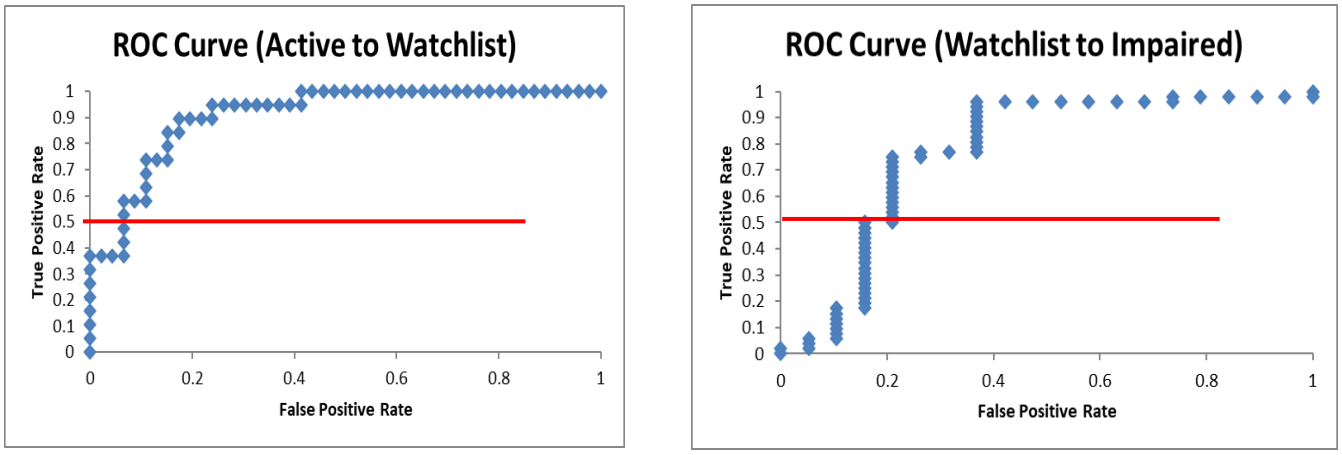


Figure 4.5

- Trade Quality Risk Score (Active - Watchlist)

Trade has total of 17 branches/scenario, it is sorted from the most safest score to the worst quality score for a customer. Value ranged from 0.019 bps to 0.604 bps. Table below shows the active quality criteria:

Nature of Account	Market Type	Payment Frequency	Corporate Status	Region	Country Rating	BNM Main Sector	Facility	Facility Currency	P-Pred
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Supplier Financing - i	MYR	0.019
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Supplier Financing - i	USD	0.022
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt	USD	0.030
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Vendor Financing Scheme	MYR	0.038
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Manufacturing	Vendor Financing Scheme - i	MYR	0.040
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	MYR	0.057
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	USD	0.067
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Supplier Financing - i	MYR	0.067
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Manufacturing	Bank Guarantee	MYR	0.079
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Import Financing - i	USD	0.087
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Revolving Credit-i	USD	0.092
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Administrative and Support Service Activities	Supplier Financing - i	MYR	0.170
Trade	Non - Traditional	Not Applicable	Non-SME	Africa	B	Manufacturing	Bank Guarantee	USD	0.192
Trade	Traditional	Not Applicable	SME-Medium	Asia	BBB+	Manufacturing	Supplier Financing - i	USD	0.266
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Bank Guarantee -i	USD	0.324
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Professional, Scientific and Technical Activities	Revolving Credit-i	MYR	0.393
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Real Estate Activities	Bank Guarantee	USD	0.604

Table 4.3

- Non-Trade Quality Risk Score (Active - Watchlist)

Non-trade has total of 26 branches/scenario, it is sorted from the most safest to the worst quality for a customer. Value ranged from 0.007 bps to 0.984 bps. Table below shows the active quality criteria:

Nature of Account	Market Type	Payment Frequency	Corporate Status	Region	Country Rating	BNM Main Sector	Facility	Facility Currency	P-Pred
Non Trade	Traditional	Monthly	Non-SME	Europe	AA-	Construction	Term Financing - i	GBP	0.007
Non Trade	Traditional	Monthly	Non-SME	Oceania	AAA	Real Estate Activities	Overseas Project Financing - i	AUD	0.033
Non Trade	Traditional	Quarterly	Non-SME	Asia	A+	Transportation and Storage	Overseas Project Financing	USD	0.060
Non Trade	Non - Traditional	Semi annual	Non-SME	Europe	B	Transportation and Storage	Overseas Investment Financing	EUR	0.064
Non Trade	Non - Traditional	Quarterly	Non-SME	Americas	BB-	Mining and Quarrying	Overseas Project Financing	USD	0.078
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB+	Manufacturing	Overseas Project Financing	USD	0.112
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Term Financing - i	MYR	0.125
Non Trade	Traditional	Quarterly	Non-SME	Asia	AAA	Administrative and Support Service Activities	Overseas Project Financing - i	USD	0.128
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Term Financing - i	USD	0.145
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB	Manufacturing	Term Financing - i	MYR	0.167
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Overseas Project Financing - i	USD	0.177
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Transportation and Storage	Term Financing - i	USD	0.193
Non Trade	Non - Traditional	Monthly	Non-SME	Asia	BBB-	Manufacturing	Contract Financing Overseas- i	USD	0.209
Non Trade	Non - Traditional	Semi annual	Non-SME	Africa	BB-	Construction	Term Loan	USD	0.246
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Manufacturing	Term Financing - i	MYR	0.349
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Transportation and Storage	Term Financing - i	MYR	0.429
Non Trade	Non - Traditional	Semi annual	Non-SME	Middle East	CCC+	Construction	Term Loan	USD	0.442
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Electricity, Gas, Steam and Air Conditioning Supply	Term Financing - i	MYR	0.520
Non Trade	Traditional	Monthly	Non-SME	Asia	BB-	Manufacturing	Overseas Project Financing - i	USD	0.544
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Term Financing - i	MYR	0.597
Non Trade	Traditional	Monthly	SME-Small	Asia	BBB+	Manufacturing	Term Financing - i	MYR	0.669
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Administrative and Support Service Activities	Overseas Investment Financing	USD	0.733
Non Trade	Non - Traditional	Monthly	SME-Medium	Africa	B-	Construction	Contract Financing Overseas- i	USD	0.904
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	CCC-	Electricity, Gas, Steam and Air Conditioning Supply	Buyer Credit	USD	0.941
Non Trade	Traditional	Monthly	SME-Small	Asia	BBB	Electricity, Gas, Steam and Air Conditioning Supply	Term Financing - i	MYR	0.956
Non Trade	Non - Traditional	Monthly	SME-Medium	Asia	Unrated	Mining and Quarrying	Term Financing - i	MYR	0.984

Table 4.4

- Trade Quality Risk Score (Watchlist - Impaired)

Trade has total of 11 branches/scenario, it is sorted by the worst score of quality customer ranged from 0.375 bps to 0.945 bps. Table below shows the impaired quality criteria:

Nature of Account	Market Type	Payment Frequency	Corporate Status	Region	Country Rating	BNM Main Sector	Facility	Facility Currency	P-Pred
Trade	Traditional	Bullet	SME-Small	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Import Financing - i	USD	0.945
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt	USD	0.943
Trade	Traditional	Bullet	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Import Financing - i	USD	0.940
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	USD	0.884
Trade	Traditional	Bullet	SME-Small	Asia	BBB+	Manufacturing	Supplier Credit	MYR	0.880
Trade	Traditional	Bullet	SME-Medium	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Letter of Credit/Trust Receipt-i	USD	0.861
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Bank Guarantee -i	USD	0.745
Trade	Traditional	Bullet	SME-Small	Asia	BBB+	Manufacturing	Letter of Credit/Trust Receipt-i	MYR	0.653
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Construction	Bank Guarantee	USD	0.566
Trade	Traditional	Not Applicable	Non-SME	Asia	BBB+	Real Estate Activities	Bank Guarantee	USD	0.462
Trade	Traditional	Not Applicable	SME-Medium	Asia	BB-	Manufacturing	Bank Guarantee	USD	0.375

Table 4.5

- Non-Trade Quality Risk Score (Watchlist - Impaired)

Non-trade has total of 43 branches/scenario, it is sorted by the worst score of quality customer ranged from 0.109 bps to 0.967 bps. Table below shows the impaired quality criteria:

Nature of Account	Market Type	Payment Frequency	Corporate Status	Region	Country Rating	BNM Main Sector	Facility	Facility Currency	P-Pred
Non Trade	Non - Traditional	Quarterly	Non-SME	Middle East	AA-	Construction	Term Financing - I	USD	0.967
Non Trade	Non - Traditional	Semi annual	Non-SME	Africa	BB-	Construction	Term Loan	USD	0.963
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	BBB-	Mining and Quarrying	Overseas Project Financing	USD	0.935
Non Trade	Traditional	Semi annual	Non-SME	Asia	BBB+	Administrative and Support Service Activities	Overseas Investment Financing	USD	0.924
Non Trade	Non - Traditional	Monthly	Non-SME	Middle East	A-	Transportation and Storage	Term Financing - I	USD	0.915
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Accommodation and Food Service Activities	Overseas Investment Financing - I	USD	0.912
Non Trade	Non - Traditional	Semi annual	Non-SME	Middle East	CCC+	Construction	Term Loan	USD	0.906
Non Trade	Non - Traditional	Monthly	SME-Medium	Asia	BBB-	Mining and Quarrying	Overseas Project Financing	USD	0.902
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB	Transportation and Storage	Overseas Investment Financing - I	USD	0.893
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Mining and Quarrying	Overseas Investment Financing	USD	0.883
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Construction	Overseas Project Financing - I	USD	0.866
Non Trade	Traditional	Quarterly	SME-Small	Asia	BBB+	Construction	Overseas Project Financing	USD	0.864
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Transportation and Storage	Buyer Credit	USD	0.863
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Construction	Buyer Credit	USD	0.854
Non Trade	Non - Traditional	Quarterly	SME-Medium	Asia	CCC	Manufacturing	Overseas Project Financing - I	USD	0.852
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB+	Transportation and Storage	Term Financing - I	USD	0.851
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Overseas Project Financing - I	USD	0.846
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Construction	Overseas Project Financing	USD	0.829
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	CCC+	Accommodation and Food Service Activities	Buyer Credit	USD	0.810
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Mining and Quarrying	Overseas Project Financing	USD	0.808
Non Trade	Non - Traditional	Monthly	SME-Small	Asia	BBB-	Accommodation and Food Service Activities	Malayalan Kitchen Financing	MYR	0.806
Non Trade	Traditional	Quarterly	Non-SME	Asia	BBB	Accommodation and Food Service Activities	Overseas Project Financing	USD	0.797
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Manufacturing	Term Financing - I	USD	0.795
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB	Transportation and Storage	Buyer Credit	USD	0.793
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	B-	Agriculture, Forestry and Fishing	Overseas Project Financing - I	USD	0.790
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Term Financing - I	USD	0.788
Non Trade	Non - Traditional	Monthly	SME-Medium	Asia	B	Professional, Scientific and Technical Activities	Overseas Project Financing - I	USD	0.769
Non Trade	Traditional	Quarterly	SME-Medium	Asia	BBB	Real Estate Activities	Overseas Project Financing	USD	0.769
Non Trade	Non - Traditional	Not Applicable	SME-Medium	Asia	B	Construction	Overseas Project Financing - I	USD	0.768
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Mining and Quarrying	Term Financing - I	USD	0.738
Non Trade	Non - Traditional	Quarterly	Non-SME	Asia	CCC-	Electricity, Gas, Steam and Air Conditioning Supply	Buyer Credit	USD	0.731
Non Trade	Traditional	Monthly	SME-Medium	Oceania	AAA	Accommodation and Food Service Activities	Malayalan Kitchen Financing	MYR	0.725
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Transportation and Storage	Import Financing - I	MYR	0.725
Non Trade	Traditional	Monthly	SME-Medium	Asia	A+	Accommodation and Food Service Activities	Malayalan Kitchen Financing	MYR	0.715
Non Trade	Traditional	Monthly	Non-SME	Asia	BBB+	Manufacturing	Term Loan	MYR	0.678
Non Trade	Non - Traditional	Monthly	Non-SME	Asia	CCC-	Construction	Overseas Project Financing	USD	0.643
Non Trade	Non - Traditional	Monthly	SME-Small	Asia	Unrated	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	MalayalanKitchen Financing Facility- I	MYR	0.630
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Manufacturing	Term Financing - I	MYR	0.478
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Transportation and Storage	Term Financing - I	MYR	0.461
Non Trade	Traditional	Monthly	SME-Medium	Asia	BBB+	Wholesale and Retail Trade, Repair of Motor Vehicles and Motorcycles	Term Financing - I	MYR	0.427
Non Trade	Traditional	Monthly	SME-Small	Asia	BBB	Electricity, Gas, Steam and Air Conditioning Supply	Term Financing - I	MYR	0.343
Non Trade	Non - Traditional	Monthly	SME-Medium	Asia	Unrated	Mining and Quarrying	Term Financing - I	MYR	0.153
Non Trade	Non - Traditional	Semi annual	Non-SME	Africa	C	Water Supply; Sewerage, Waste Management and Remediation Activities	Buyer Credit	EUR	0.109

Table 4.6

Active Criteria Score (Trade)

Figures below shows sensitivity of 10 branches/scenario out of 17 branches link with the safest **score for trade to remain active/stage 1**:

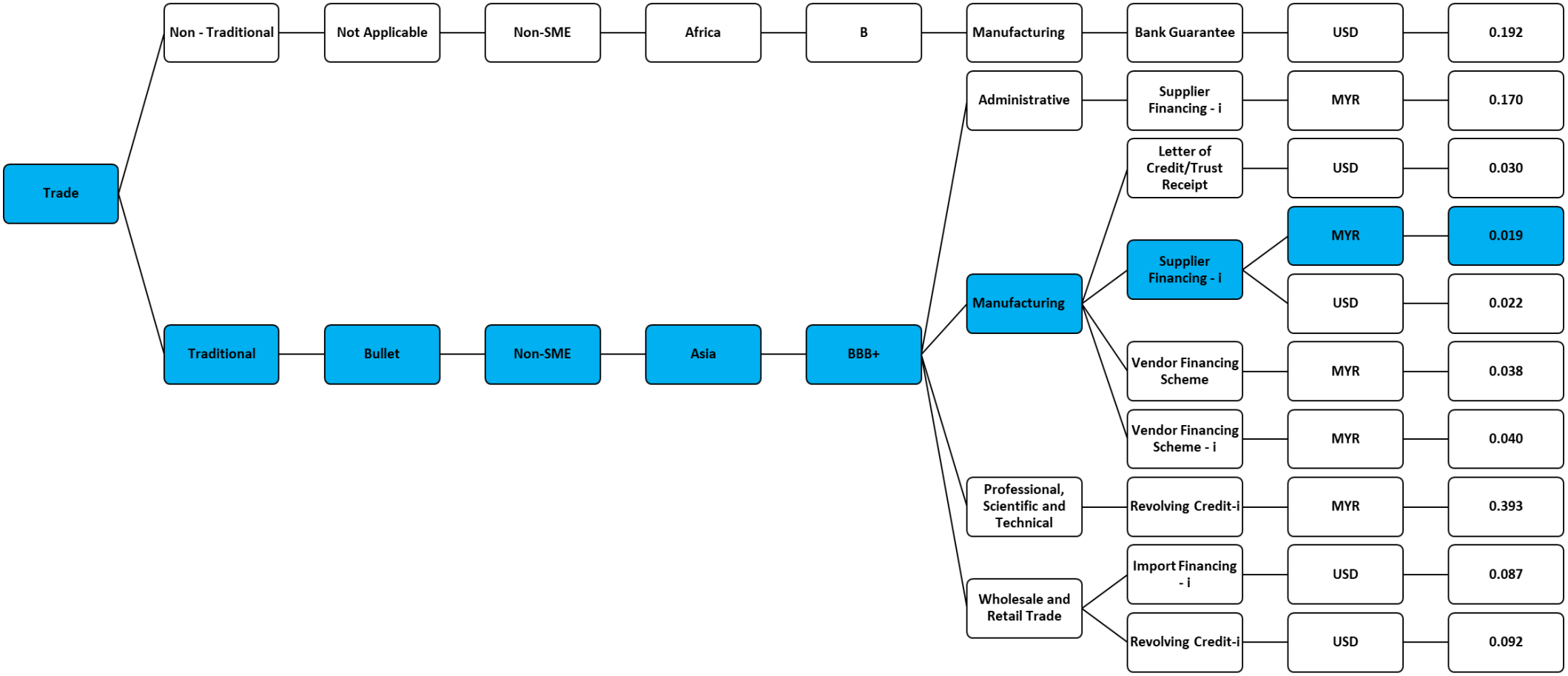


Figure 5.0

Active Criteria Score (Non-Trade)

Figures below shows sensitivity of 10 branches/scenario out of 26 branches link with the safest score for **non-trade to remain active/stage 1:**

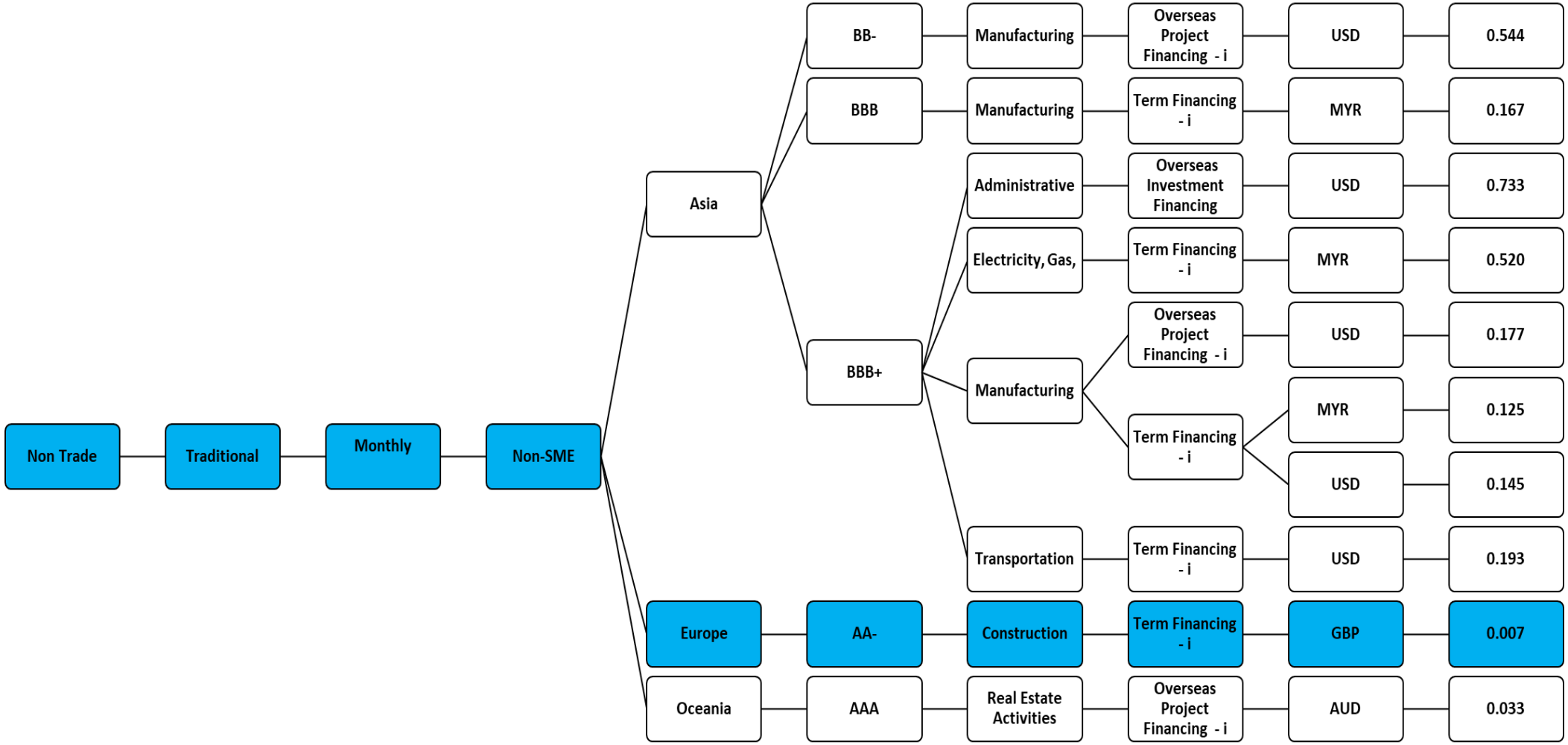


Figure 5.1

Impaired Criteria Score (Trade)

Figures below shows sensitivity of 11 branches/scenario out of 11 branches link with the most concern score for **trade to become impaired/stage 3**:

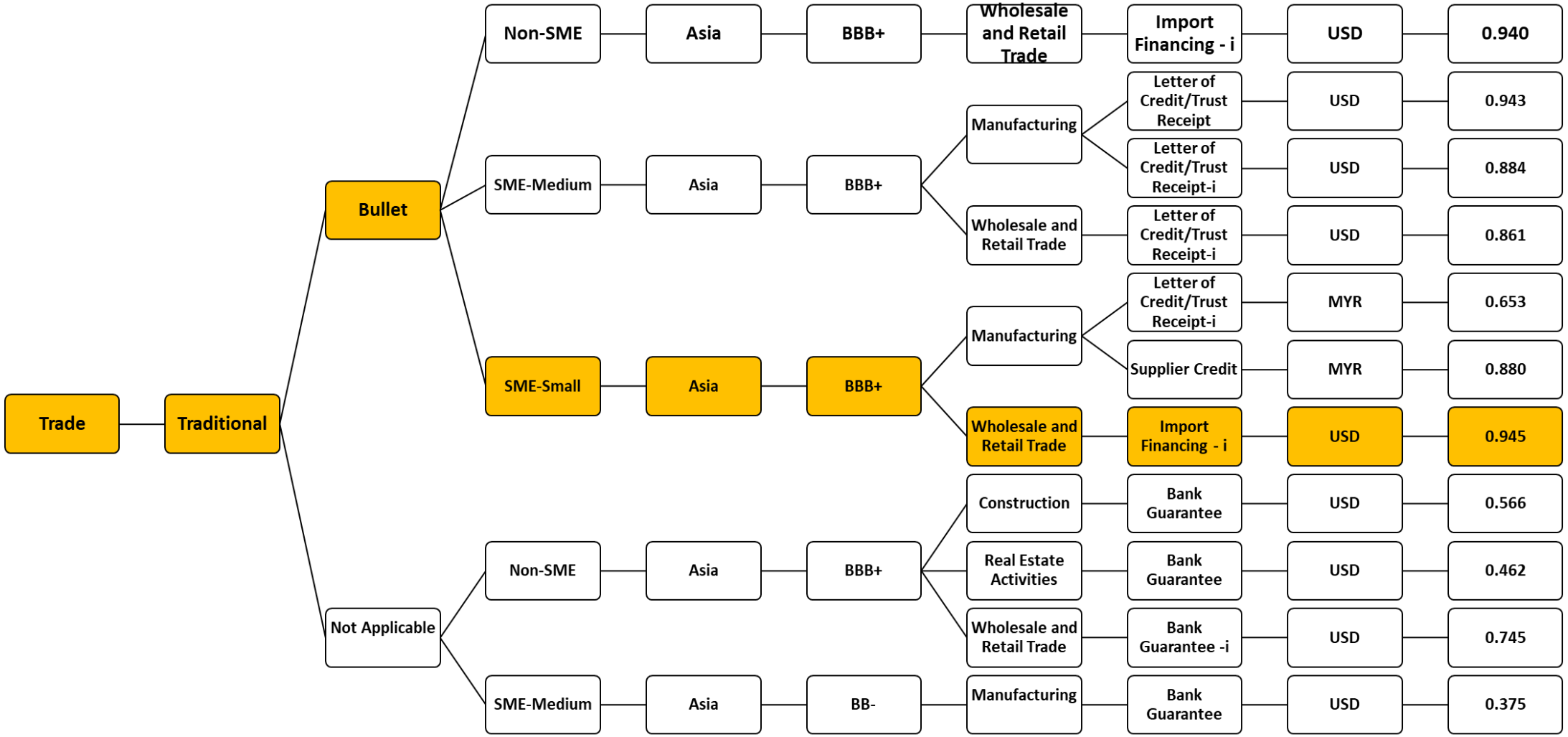


Figure 5.2

Impaired Criteria Score (Non-Trade)

Figures below shows sensitivity of 7 branches/scenario out of 59 branches link with the most concern score for **trade to become impaired/stage 3**:

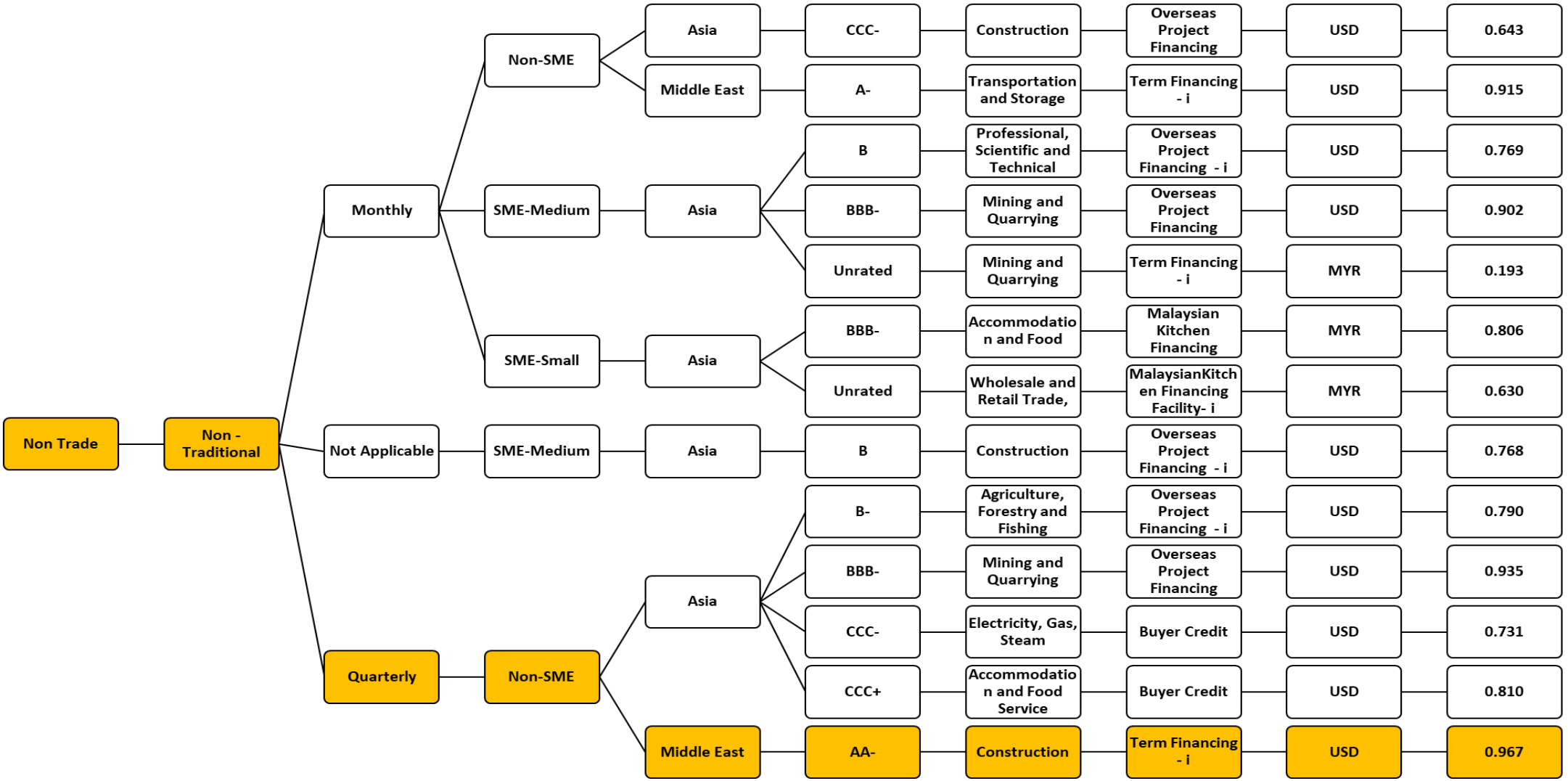


Figure 5.3

Research Question:

"Which of Trade and Non-Trade is more quality in term of profit and financing asset? Which is risker to do business and where could EXIM Bank focus to expand for more profit? How does EXIM Bank perform complying the mandated role?"

Key Takeaway:

"EXIM Bank of Malaysia has fulfilled its mandate by providing credit facilities to finance and support exports and imports of goods, services, and overseas projects to 26 different countries with 20 different facility and 5 different currencies across different sectors and entities. With non-trade being 90% of its financing asset, **trade** is more secure whereby 10% of trade is making 3.82% of profit compared to non-trade only at 2.95%. As well as the countries EXIM Bank has ventured into, the underserved countries such as **Thailand and Eswatini** are more secure compared to UAE, Saudi and many more. The research also found sector in **Electricity, Gas, Steam and Air Conditioning Services** has more secure profit by 6.29% compared to the highest amounted loan, Mining and Quarrying by 2.99%. These high secure profits not only show **good performance in getting profit** but rather in term of the **customer quality** as well, it shows good indicator in payment/repayment for trade customer, customer expose in Thailand and Eswatini, and even customer venture into Electricity, Gas, Steam and Air Conditioning Services. Therefore, believing that our focus in profit security will pay off, a better profit security for the bank to generate profit will gives confident for the bank to recognise its long-term growth strategy."

Appendix

The following figure lists the **top 15 customers** in terms of **largest loan**:

Customer Name	Country Exposure	Asset Quality Status	Facility	Payment Frequency	Loans Outstanding	Accrued Profit	Profit Payment
Yinson Boronia Production B.V.	Brazil	Active	Overseas Project Financing	Quarterly	RM563,800,306.92	RM8,192,434.92	RM32,547,638.08
PT Sriwijaya Air	Indonesia	Impaired	Buyer Credit	Monthly	RM380,100,012.14	RM0.00	RM0.00
Markmore Energy (Labuan) Limited	Kazakhstan	Impaired	Overseas Project Financing	Quarterly	RM378,948,389.76	RM0.00	RM0.00
The Ministry of Finance Government of Lao PDR	Lao People's Democratic Republic	Active-Watchlist	Buyer Credit	Quarterly	RM248,695,532.61	RM2,952,032.82	RM15,699,340.98
Felda Global Ventures Capital Sdn Bhd	Malaysia	Active	Supplier Financing - i	Bullet	RM240,376,999.32	RM376,949.32	RM8,211,204.09
OM Materials (Sarawak) Sdn Bhd	Malaysia	Active	Overseas Project Financing	Quarterly	RM226,679,439.39	RM105,596.26	RM14,638,169.08
S P Setia Berhad	United Kingdom	Active	Term Financing - i	Bullet	RM202,193,262.63	RM831,741.14	RM11,964,285.12
HCD-PRBSB Co. Ltd.	Cambodia	Impaired	Overseas Project Financing - i	Not Applicable	RM187,208,861.80	RM0.00	RM0.00
Istanbul Sabiha Gokcen Uluslararasi Havalimani Yatirim Yapim Isletme A.S	Turkiye	Active	Overseas Investment Financing	Semi annual	RM154,741,038.43	RM2,825,334.02	RM4,787,450.98
Eversendai Corporation Berhad	United Arab Emirates	Impaired	Term Financing - i	Semi annual	RM140,763,666.60	-RM0.00	RM0.00
INGRESS INDUSTRIAL (MALAYSIA) SDN BHD	Indonesia	Active	Term Financing - i	Monthly	RM131,340,371.95	RM63,013.12	RM5,840,955.34
Al Manara Water Company Limited	Sudan	Impaired	Buyer Credit	Semi annual	RM104,422,553.53	RM0.00	RM933,163.88
Impiana Ubud (Labuan) Co. Ltd (formerly known as Impiana Investment Co. Ltd.)	Indonesia	Impaired	Overseas Project Financing	Quarterly	RM103,128,814.05	RM0.00	RM0.00
Superboom Projects Sdn Bhd	Indonesia	Impaired	Overseas Project Financing	Bullet	RM93,694,364.03	RM0.00	RM0.00
Hummingbird Energy (L) Inc	Indonesia	Impaired	Overseas Investment Financing	Quarterly	RM89,229,815.47	RM0.00	RM0.00

Table 5.0

The following figure lists the **top 15 impaired customers** in terms of **largest loan**:

Customer Name	Country Exposure	Facility	Payment Frequency	Loans Outstanding	Accrued Profit	Profit Payment	Year
PT Sriwijaya Air	Indonesia	Buyer Credit	Monthly	RM380,100,012.14	RM0.00	RM0.00	2018
Markmore Energy (Labuan) Limited	Kazakhstan	Overseas Project Financing	Quarterly	RM378,948,389.76	RM0.00	RM0.00	2015
HCD-PRBSB Co. Ltd.	Cambodia	Overseas Project Financing - i	Not Applicable	RM187,208,861.80	RM0.00	RM0.00	2019
Eversendai Corporation Berhad	United Arab Emirates	Term Financing - i	Semi annual	RM140,763,666.60	-RM0.00	RM0.00	2022
Al Manara Water Company Limited	Sudan	Buyer Credit	Semi annual	RM104,422,553.53	RM0.00	RM933,163.88	2016
Impiana Ubud (Labuan) Co. Ltd (formerly known as Impiana Investment Co. Ltd.)	Indonesia	Overseas Project Financing	Quarterly	RM103,128,814.05	RM0.00	RM0.00	2021
Superboom Projects Sdn Bhd	Indonesia	Overseas Project Financing	Bullet	RM93,694,364.03	RM0.00	RM0.00	2021
Hummingbird Energy (L) Inc	Indonesia	Overseas Investment Financing	Quarterly	RM89,229,815.47	RM0.00	RM0.00	2019
Sapura TMC Sdn Bhd	Malaysia	Overseas Investment Financing	Semi annual	RM86,752,965.60	RM0.00	RM3,764.97	2022
PT Sari Indah Lestari	Indonesia	Buyer Credit	Quarterly	RM84,151,740.74	RM0.00	RM0.00	2009
Daya Maritime Limited	Malaysia	Term Financing - i	Semi annual	RM56,335,695.07	RM0.00	RM0.00	2016
Kota Laksamana 101 Limited	Malaysia (Labuan)	Term Financing - i	Monthly	RM53,545,329.20	RM0.00	RM0.00	2018
Hondaafushi Investment Private Limited	Maldives	Buyer Credit	Quarterly	RM43,805,701.58	RM0.00	RM0.00	2013
UPL Co. Ltd	Lao People's Democratic Republic	Overseas Project Financing	Monthly	RM27,982,730.28	RM0.00	RM0.00	2018
Hyrax Oil Sdn Bhd	Sri Lanka	Overseas Project Financing - i	Quarterly	RM18,376,850.14	RM0.00	RM1,241,093.67	2021

Figure 5.1

The following figure lists the **top 15 impaired customers** in terms of **longest impaired**:

Customer Name	Country Exposure	Facility	Payment Frequency	Loans Outstanding	Accrued Profit	Profit Payment	Year In
PT Eden Capital Indonesia	Indonesia	Overseas Project Financing	Quarterly	RM16,022,908.03	RM0.00	RM0.00	2008
PT Sari Indah Lestari	Indonesia	Buyer Credit	Quarterly	RM84,151,740.74	RM0.00	RM0.00	2009
PT Bumi International Tankers	Indonesia	Buyer Credit	Quarterly	RM13,428,030.74	RM0.00	RM0.00	2010
PT Layar Sentosa Shipping Corporation	Indonesia	Buyer Credit	Quarterly	RM13,623,590.22	RM0.00	RM0.00	2010
PT Mahakarya Inti Buana	Indonesia	Buyer Credit	Quarterly	RM13,487,764.64	RM0.00	RM0.00	2010
Solartif Sdn Bhd	Malaysia	Supplier Credit	Bullet	RM637,264.90	RM0.00	RM121,014.32	2012
Hondaafushi Investment Private Limited	Maldives	Buyer Credit	Quarterly	RM43,805,701.58	RM0.00	RM0.00	2013
Nirmala Villas Restaurant Sdn Bhd	India	Malaysian Kitchen Financing	Monthly	RM794,049.34	RM0.00	RM0.00	2013
Virtual Peak Group Sdn Bhd	Malaysia	Overseas Project Financing	Quarterly	RM8,133,918.01	RM0.00	RM0.00	2013
Eaglexpress Air Charter Sdn Bhd	Saudi Arabia	Term Financing - i	Monthly	RM9,158,961.56	RM0.00	RM207,237.13	2015
International Halal Meat Exporter LLC	Mongolia	Overseas Project Financing - i	Quarterly	RM13,105,040.71	RM0.00	RM0.00	2015
Markmore Energy (Labuan) Limited	Kazakhstan	Overseas Project Financing	Quarterly	RM378,948,389.76	RM0.00	RM0.00	2015
3 Plus 1 Studio Sdn Bhd	China	Malaysian Kitchen Financing	Monthly	RM1,000,848.23	RM0.00	RM0.00	2016
Al Manara Water Company Limited	Sudan	Buyer Credit	Semi annual	RM104,422,553.53	RM0.00	RM933,163.88	2016
Daya Maritime Limited	Malaysia	Term Financing - i	Semi annual	RM56,335,695.07	RM0.00	RM0.00	2016
PT Great Colour Energy	Indonesia	Overseas Project Financing - i	Quarterly	RM16,050,046.28	RM0.00	RM4,500.00	2016

Figure 5.2

The following table Shows the **country fitch rating** by its region and country exposure:

Country Rating	Region	Country Exposure
AAA	Asia	Singapore
AAA	Europe	Germany
AAA	Oceania	Australia
AA+	Americas	Canada
AA+	Americas	United States
AA+	Oceania	New Zealand
AA-	Asia	Chinese Taipei / Taiwan
AA-	Asia	Korea, Republic of (South)
AA-	Europe	United Kingdom
AA-	Middle East	United Arab Emirates
A+	Asia	China
A	Asia	Japan
A-	Middle East	Saudi Arabia
BBB+	Asia	Malaysia
BBB+	Asia	Malaysia (Labuan)
BBB+	Asia	Thailand
BBB+	Europe	Spain
BBB	Asia	Indonesia
BBB-	Asia	India
BBB-	Asia	Kazakhstan
BB-	Africa	Seychelles
BB-	Americas	Brazil
BB-	Asia	Bangladesh
BB-	Asia	Vietnam
B	Africa	Tanzania, United Republic of
B	Asia	Cambodia
B	Europe	Türkiye
B-	Africa	Eswatini
B-	Africa	Ghana
B-	Asia	Mongolia
B-	Europe	Bosnia and Herzegovina
CCC+	Asia	Maldives
CCC+	Middle East	Iraq
CCC	Asia	Sri Lanka
CCC-	Asia	Lao People's Democratic Republic
C	Africa	Sudan
Unrated	Asia	Brunei Darussalam

Table 2.0